

SOCIAL PROTECTION AND LABOR

DISCUSSION PAPER

No. 2604 | September 2025

The heat is on: How can long-term care systems in Europe and Central Asia promote climate adaptation?

Sarah Coll-Black
Ali Hamandi
Aaron Beitman
Andrey Tretyak
Valeria Arias Salvador

© 2025 International Bank for Reconstruction and Development / The World Bank

1818 H Street NW, Washington DC 20433

Telephone: +1 (202) 473 1000; Internet: www.worldbank.org.

This work is a product of the staff of The World Bank with external contributions. The findings, interpretations, and conclusions expressed in this work do not necessarily reflect the views of The World Bank, its Board of Executive Directors, or the governments they represent.

The World Bank does not guarantee the accuracy of the data included in this work. The boundaries, colors, denominations, and other information shown on any map in this work do not imply any judgment on the part of The World Bank concerning the legal status of any territory or the endorsement or acceptance of such boundaries.

RIGHTS AND PERMISSIONS

The material in this work is subject to copyright. Because The World Bank encourages dissemination of its knowledge, this work may be reproduced, in whole or in part, for noncommercial purposes as long as full attribution to this work is given.

Any queries on rights and licenses, including subsidiary rights, should be addressed to World Bank Publications, The World Bank Group, 1818 H Street NW, Washington, DC 20433, USA; fax: +1 (202) 522 2625; e-mail: pubrights@worldbank.org.

The heat is on: How can long-term care systems in Europe and Central Asia promote climate adaptation?

Sarah Coll-Black, Ali Hamandi, Aaron Beitman, Andrey Tretyak and Valeria Arias Salvador



Table of Contents

Acknowledgements.....	4
Acronyms	5
Executive Summary.....	6
Section I: Introduction.....	7
Section II: Definitions, frameworks, and methodology.....	9
Section III: How is climate change affecting people with LTC needs?	12
Section IIIa: Hazard, exposure, vulnerability, and response framework.....	13
Section IIIb: Climate hazards in ECA.....	14
Section IIIc: Vulnerability to climate hazards	16
Section IIId: Direct impacts of climate hazards on health and functional ability.....	17
Section IIIe: LTC system impacts.....	20
Section IV: LTC systems responses to the impact of climate hazards	22
Section IVa: United Kingdom.....	23
Section IVb: Spain.....	25
Section IVc: France	28
Section IVd: Japan	30
Section IVe: United States	31
Section IVf: Learning from selected OECD interventions.....	34
Section V: Adapting LTC systems to climate change	36
Section Va: Before climate hazard events	36
Section Vb: During climate hazards events	39
Section Vc: After climate hazard events.....	40
Section VI: What is the scope for future work?	41
Section VII: Conclusion.....	43
References.....	44

List of Tables

Table 1: LTC system functions.....	12
Table 2: Heat-Health Alerting System levels and impacts.....	23
Table 3: UKHSA cold weather response guidance for social care sector managers.....	25
Table 4: Community of Madrid heat risk levels and impacts.....	27
Table 5: Community of Madrid Intervention plan response measures in LTC facilities: Red Alert	27
Table 6: Legal requirements related to long-term care in France’s National Heatwave Plan	29
Table 7: Japan’s heat stroke alert - heat illness measures	30
Table 8: Extreme Heat and Poor Air Quality Events in the Climate Hazard Adaptation Profiles of the Massachusetts Department of Public Health (United States) LTC-related Measure	32
Table 9: Selected OECD country climate hazard interventions	35

List of Figures

Figure 1: Interaction between determinants of climate risk	13
Figure 2: Direct impacts of climate hazards on health and functional ability, and their implications for LTC systems.....	20
Figure 3: Climate hazard impacts on LTC systems.....	21
Figure 4: Sheltering, relocation, and evacuation decision tree.....	33
Figure 5: Measures and interventions to strengthen LTC systems before, during, and after climate hazard events.....	41

List of Boxes

Box 1: Long-Term Care in Europe and Central Asia: a brief overview.....	8
Box 2: Defining key terms in long-term care and climate change.....	9
Box 3: Summary of key climate hazards in ECA	16
Box 4: UKHSA Heat-Health Alerting long-term preparedness	24
Box 5: Evidence from Europe points to the benefits of harnessing LTC systems for climate response	36

Acknowledgements

This working paper is a joint product of the Health, Nutrition, and Population (HNP) and the Social Protection and Labor (SPL) Global Practices of the World Bank. The paper was written by Sarah Coll-Black, Ali Hamandi, Aaron Beitman, Andrey Tretyak, and Valeria Arias Salvador. Stephen Dorey (Senior Health and Environment Specialist, Health, Nutrition & Population Global Practice, World Bank) and Solene Rougeaux (Senior Social Protection Specialist, Europe and Central Asia Region, World Bank) provided peer review comments. Dr. Lorraine Frisina (Tenured Lecturer in Public Health at the University of Bremen), and Fiona Stewart (Lead Economist and Global Lead on Pensions and Aging, Social Protection and Labor Global Practice, World Bank) provided valuable comments on the draft. The working paper benefited from discussions with Professor Shereen Hussein (London School of Hygiene and Tropical Medicine), who also provided input into an earlier draft. Indhira Santos (Lead Economist, Human Development, Europe and Central Asia Region, World Bank) provided overall guidance together with Paolo Belli (Practice Manager, Social Protection and Labor Global Practice, Europe and Central Asia Region, World Bank).

The analysis was made possible due to the generous funding from the World Bank's Climate Support Facility (CSF). The mission of the CSF is to support developing countries in accelerating their transition to low-carbon and climate-resilient development and elevate the national decarbonization agenda.

Acronyms

ADL	Activities of Daily Living
AEMET	National Meteorological Agency (Spain)
AI	Artificial Intelligence
ASP	Adaptive Social Protection
BCP	Business Continuity Plans
C	Celsius
CCDR	Country Climate and Development Reports
CHA	Cold-Health Alert
CHAP	Climate Hazard Adaptation Profiles
ECA	Europe and Central Asia
EPA	Environmental Protection Agency (United States)
EU	European Union
HCBS	Home- and Community-Based Services
HHA	Heat-Health Alert
HSA	Heat Stroke Alert (Japan)
IPCC	Intergovernmental Panel on Climate Change
LTC	Long-Term Care
OECD	Organisation for Economic Co-Operation and Development
SACS	Heatwave and Health Alert System (France)
UKHSA	United Kingdom Health Security Agency

Executive Summary

Like in many regions, countries in the World Bank's Europe and Central Asia (ECA) region are simultaneously experiencing population aging and increasing climate hazards, albeit with local variation in prevalence, intensity, and frequency. As populations age, the functional ability of people declines, resulting in rising needs for long-term care (LTC). At the same time, older people and those with care needs are particularly vulnerable to the impacts of climate change. Strengthening the provision of LTC, defined as the services and support that aid people in developing and maintaining their functional abilities, can therefore play a critical role in promoting the climate resilience of people with care needs, given their enhanced vulnerability to climate change.

Drawing on case studies from Organisation for Economic Co-Operation and Development (OECD) countries and a broader review of the literature, this working paper identifies a set of measures that spans the functions of an LTC system, which ECA countries could adopt before, during, and after a climate hazard event. These measures aim to maintain continuity of LTC services and support, respond to climate-induced needs among LTC beneficiaries, and reach people affected by climate change who are outside the LTC system and for whom the LTC system may be an appropriate response modality. These involve modifications to the formal and informal provision of LTC, while also promoting actions that individuals and their caregivers can take to protect themselves. Many of these actions involve adapting how LTC services are organized and delivered—through preparedness protocols, workforce training, facility standards, and coordination mechanisms—rather than creating new, standalone services.

The analysis points to several key policy messages:

- Older people and those with functional limitations are disproportionately affected by climate hazards, making LTC systems a natural platform for climate adaptation.
- Climate resilience in LTC does not necessarily require the development or expansion of new services and support but the adaptation of existing ones, ensuring preparedness and continuity.
- Measures must be timed and sequenced—before, during, and after hazard events—to reduce disruption, safeguard lives, and reach vulnerable populations both within and outside formal LTC systems.
- OECD country experiences, such as weather alerting systems tied to LTC responses, registries of vulnerable individuals, and facility adaptation standards, demonstrate practical models that can be tailored to ECA contexts.
- Embedding climate adaptation into LTC reforms is urgent. Doing so provides a dual benefit: stronger systems to meet rising care needs from demographic change and greater resilience to the climate hazards that increasingly shape the region's future.

The intersection of rapid demographic aging, rising functional limitations, and the increasing prevalence, intensity, and frequency of climate hazards in the ECA region underscores the urgent need for LTC systems to integrate climate adaptation measures into their design, planning, and provision of services and support.

Section I: Introduction

While aging societies are a feature of many countries, the Europe and Central Asia (ECA) region has some of the world's oldest populations. The pace and drivers of aging vary across the region, however. In the European Union (EU), population aging has been unfolding for decades—primarily due to sustained low fertility and increasing longevity. In contrast, Türkiye and countries in the South Caucasus, Central Asia, and parts of the Western Balkans still have relatively young populations but are aging rapidly due to falling birth rates, rising life expectancy, and, in some cases, high levels of emigration (World Bank forthcoming; WHO 2020; OECD 2025). Across the region, these demographic trends are closely linked to a rising need for long-term care (LTC), as aging populations face higher rates of functional decline (Box 1).

Climate hazards in ECA are increasing in frequency and intensity, posing uneven risks to populations across the region. The Intergovernmental Panel on Climate Change (IPCC), the United Nations body for assessing the science related to climate change, expects that climate hazards will intensify and become more recurrent in the coming decades (IPCC 2023). In ECA countries, extreme heat and cold waves, floods, landslides, wildfires, water stress, and droughts have been identified as significant climate hazards, although their prevalence, intensity, and frequency depend on local conditions. In areas exposed to climate hazards, impacts on populations are uneven, as the vulnerability of individuals determines their susceptibility. Two examples from Europe illustrate this disproportionality. A European heatwave in 2022 resulted in about 62,000 deaths, with roughly half among individuals 80 and older (Ballester et al. 2023). Catastrophic flooding in Spain's Valencia region in 2024 led to more than 200 deaths, about half of which were people older than 70 (AGE 2024). Additionally, the mortality rate for people with disabilities during natural disasters is estimated to be four times higher than that for people without disabilities (UNESCAP 2019).

LTC systems provide a crucial entry point for protecting people with functional dependency from the impacts of climate change. LTC systems are designed to support people with functional limitations, defined as the capacity to perform basic activities necessary for daily living (e.g., mobility, self-care, and communication) that are essential for maintaining independence and quality of life. These limitations often increase with age but can affect individuals at any stage of life, arising from disabilities and chronic health conditions (WHO 2015). Growing evidence shows biological and health, social, and systemic and structural factors increase the vulnerability of older people and people with disabilities to climate hazards, populations which are most likely to experience functional limitations. Given their goal of supporting people with functional limitations due to age-related frailty, disability, and/or chronic health conditions, LTC systems can be an efficient pathway to protect these populations from climate change, thereby promoting their climate adaptation.

Though there is no consensus definition of LTC, there is general agreement on its contours. LTC includes a range of health and social services and support provided over extended or indefinite periods to individuals with chronic health conditions, disability,¹ or age-related frailty (WHO 2015). In contrast with

¹ Not all people living with disabilities will need LTC services and support and, among those who do, they may also require tailored

acute care, LTC does not equate to rehabilitative or palliative care, although there are overlaps. LTC is typically performed by formal and/or informal caregivers² and delivered in a variety of settings, such as private homes, communities, or residential facilities. LTC is provided directly to people in the form of services, which may be offered by government agencies, contracted out to or purchased directly from private providers, or provided indirectly through government-provided cash benefits or vouchers, which are then used by beneficiaries or their families to purchase services and support. LTC systems are not necessarily standalone systems but rather constellations of services and support offered through the health and social protection sectors, each with its own governance models, financing mechanisms, care provision approaches, and degrees of involvement from various levels of government and the private sector.³

Box 1: Long-Term Care in Europe and Central Asia: a brief overview

In ECA, LTC systems vary in structure: while some countries have relatively unified systems, most feature services and support delivered through the health and social protection sectors. This often leads to fragmented governance and financing, involving multiple agencies across levels of government and creating administrative complexities and gaps in service delivery.⁴ At the same time, limited regulatory oversight of private providers complicates efforts to ensure quality and accountability. Formal LTC provision in ECA tends to be limited in coverage and scope, with financing structures relying on a mix of public budgets and private contributions that often lack coherence. As a result, informal care—primarily provided by family members, especially women—remains the cornerstone of LTC provision in the region, carrying significant implications for caregivers’ economic participation and overall well-being.

In parts of ECA, residential facilities and hospital infrastructure have historically played a central role in providing LTC, reflecting a lack of development in alternative care options.⁵ However, the expansion of home- and community-based services (HCBS), particularly for individuals with low to moderate care needs, has gained attention in policy discussions. HCBS can help prevent avoidable hospitalizations and serve as a complement to residential care providing a range of services and support that can accommodate different needs, especially as people age.⁶

Source: World Bank (forthcoming); Spasova et al. (2018); Sinnott et al. (2015); World Bank (2010).

disability services to promote inclusion, education and employment outcomes. The exact division between LTC and care and support for people with disabilities is nuanced and varies by country.

² The term informal caregiver, also referred to as informal carer, refers to an individual who provides care (typically unpaid) to an older or dependent person within the context of a social relationship—such as a spouse, family member, friend, neighbor, or other non-kin. This care may include assistance with household tasks, transportation, emotional support, social companionship, or coordination of professional services (Triantafyllou et al., 2010).

³ See World Bank (forthcoming) for a discussion.

⁴ There are often different rules and procedures for people with functional limits arising from age, chronic health conditions or disabilities to access LTC services and support, with potentially different co-payment mechanisms as well, which contribute to further fragmentation.

⁵ Many ECA countries are characterized by their previous overreliance on large residential facilities to provide care for people with disabilities, including their care needs. Nowadays, the independent living and deinstitutionalization agenda promotes home and community-based care through a dedicated effort towards deinstitutionalization.

Source: [https://www.europarl.europa.eu/RegData/etudes/STUD/2024/754233/IPOL_STU\(2024\)754233_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/STUD/2024/754233/IPOL_STU(2024)754233_EN.pdf)

⁶ In addition, the traditional focus of LTC in ECA has been to address the physical needs of recipients, but as the prevalence of dementia soars, a more holistic provision of care is needed (Warsaw and Bragg 2014).

This working paper explores strategies and interventions that LTC systems in ECA could adopt to strengthen the climate resilience of people with functional limitations and their caregivers.⁷ After a short section on methodology and conceptual framing, the paper introduces the IPCC climate risk framework and outlines the main climate hazards projected for ECA, noting their varying frequency and intensity across the region. It then examines how individual vulnerability shapes the impact of these hazards, with a focus on older people and people with disabilities, whose functional limitations and care needs increase their susceptibility. The analysis considers biological and health factors, social circumstances, and structural conditions that heighten vulnerability to both climate shocks and gradual changes such as rising temperatures. The paper then turns to international experience, highlighting approaches from selected OECD countries that have modified their LTC systems or introduced innovations to protect people with care needs—and, in some cases, a broader population of older people and people with disabilities—from climate risks. Building on these case studies and a wider literature review, the paper identifies a set of strategies across LTC system functions that ECA countries could apply before, during, and after climate hazards to maintain essential care, address emerging needs, and extend support to groups not currently covered. The paper concludes with questions for further research and analysis.

Section II: Definitions, frameworks, and methodology

This section describes the methodology used to collect and analyze the information presented in the sections that follow. This includes the key definitions used (Box 2), the systematic literature review conducted, the selection of country case studies presented, and the framework used to analyze LTC systems.

Box 2: Defining key terms in long-term care and climate change

The concepts and definitions used in this paper are presented here to aid the reader, given the ensuing discussion spans the health, social protection, and climate change sectors. Sources are listed in places to facilitate further exploration, by interested readers.

- *Older people* include individuals above the age of 65 years.
- *People living with disabilities* include individuals who have long-term physical, mental, intellectual, or sensory impairments which in interaction with various barriers may hinder their full and effective participation in society on an equal basis with others (UNCRPD 2006).
- *Long-term care (LTC)* is a range of services and support for people who, as a result of mental and/or physical frailty, chronic conditions, and/or disability over an extended period, depend on help with daily living activities and/or need permanent nursing care. The daily living activities for which help is needed may be the self-care activities that a person must perform every day (ADLs) or may be related to independent living (IADLs) (EU 2014).
- *Activities of daily living (ADLs)* refer to basic self-care tasks, such as bathing, dressing, eating, and toileting.

⁷ This working paper focuses on the potential role of LTC systems in promoting the climate adaptation. It refers to but does not fully consider the implications of climate change on the need for health care or broader adaptation measures in urban planning or other sectors to protect vulnerable populations from climate hazards.

- *Instrumental activities of daily living (IADL)* refer to more complex activities that allow an individual to live independently in a community, such as managing finances, shopping, and cooking.
- *LTC facilities* are institutions where LTC is formally provided, such as nursing or care homes, assisted living residences, or rehabilitation centers.
- *Community-based care* is LTC services and support provided in community, such as in senior citizen centers or adult day centers. These services and support can promote aging in place (WHO 2015).
- *Climate adaptation* means adjustment in ecological, social, or economic systems in response to actual or expected climatic stimuli and their effects or impacts. It refers to changes in processes, practices, and structures to moderate potential damages or to benefit from opportunities associated with climate change (IPCC 2023).
- *Climate hazard* is the potential occurrence of a natural or human-induced physical event or trend that may cause loss of life, injury, or other health impacts as well as damage and loss to property, infrastructure, livelihoods, service provision, ecosystems, and environmental resources (IPCC 2023).
- *Climate resilience* can be understood as the ability of a system and its components to anticipate, absorb, accommodate, or recover from the effects of a hazardous event in a timely and efficient manner, including through ensuring the preservation, restoration, or improvement of its essential basic structures and functions (Cardona et al. 2012).
- *Exposure* is the presence of people; livelihoods; species or ecosystems; environmental functions, services, and resources; infrastructure; or economic, social, or cultural assets in places or settings that could be adversely affected by climate-related hazards (IPCC 2023).
- *Extreme weather event* refers to an event that is rare at a particular place and time of year, with unusual characteristics in terms of magnitude, location, timing, or extent (WMO 2025).
- *Slow onset events* refer to the risks and impacts associated with increasing temperatures, desertification, loss of biodiversity, land and forest degradation, glacial retreat and related impacts, ocean acidification, sea level rise, and salinization (UNFCCC 2012).
- *Vulnerability* is the propensity or predisposition to be adversely affected; vulnerability encompasses a variety of concepts and elements, including sensitivity or susceptibility to harm and lack of capacity to cope and adapt (Cardona et al. 2012).

Source: Authors, based on listed sources (2025).

To analyze the vulnerability to climate change of people with care needs, this paper considers evidence on how climate change affects older people and people with disabilities. Evidence on the exposure and vulnerability of these two populations is used as a proxy for assessing that of people with LTC needs, given that functional limitations rise steeply with age and that disability and care needs strongly correlate (WHO 2015). This paper focuses on evidence of increased care needs or impacts that are strongly associated with functional limitations and/or could be directly addressed through LTC services and support.

Methodologically, the paper draws on multiple sources. A literature review of published sources, including World Bank climate data sources and the Country Climate and Development Reports (CCDRs),⁸ informs the analysis of climate hazards in ECA as well as the exposure and vulnerability of older people and people with disabilities. A systematic literature review of articles published in English—using various combinations of search terms—informs the analysis of the impact of climate hazards on older people and people with disabilities and on LTC services and support that are provided in different settings. The paper analyzes publications on LTC and climate adaptation, emergency preparedness and response, and risk

reduction and response. It also includes peer reviewed articles, policy analysis, and publications from national and international organizations.

The case studies in section five illustrate responses to different climate hazards through various facets of the LTC system. This paper selected case studies from the United Kingdom, Spain, France, Japan, and the United States because information on their LTC systems is well documented. In addition, they represent a diverse set of institutional arrangements, financing mechanisms, and policy responses to demographic pressures and now climate change.

- **United Kingdom** offers insights from a universal, tax-funded system that is integrating climate adaptation into LTC policies at the national and local levels.
- **Spain** shows the evolution of a largely decentralized LTC system to include climate adaptation measures at both the national and local levels.
- **France** reveals how to combine public funding with a strong emphasis on in-home care and introduce adaptation measures for older adults as part of a national climate strategy.
- **Japan**, one of the most rapidly aging societies in the world with a long-established social insurance-based LTC system, illustrates the integration of people with care needs into disaster preparedness and response.
- **United States**, with a largely mixed public-private system, provides examples of localized approaches, public health emergency planning, and targeted interventions to reduce vulnerability to climate change among older people and people living with disabilities.

The case studies aim to illustrate practical strategies that could be adapted or piloted in ECA countries, considering local governance structures, fiscal space, and service delivery models. These case studies do not reflect a systematic review of climate interventions focused on people with care needs nor the complete LTC systems in these countries. Instead, the examples aim to show how selected OECD countries are responding to the impacts of climate hazards on people with care needs specifically, with direct relevance to LTC systems.

This working paper uses a framework developed by the World Bank to describe the functions of an LTC system.⁸ The framework, as presented in Table 1, helps categorize existing interventions in the case studies surveyed and the recommended measures aimed at reducing the impact of climate hazards on people with care needs. The framework includes the core functions of governance, financing, human resources, and the delivery of services and support. These functions are interdependent, collectively contributing to the mobilization of resources, the delivery of services and support, and the achievement of improved health and well-being outcomes. Strengthening one function, such as governance, can reinforce others, while weaknesses in any area can compromise overall system performance. This underscores the need for a coordinated, system-wide approach to LTC development, particularly when addressing complex challenges like climate hazards that require resilience and adaptability across all system functions.

Table 1: LTC system functions

Governance	Financing	Human resources	LTC services and support
Formulating, implementing, and overseeing policy across all other system functions. This entails political leadership and stewardship of the system and establishment of institutional structures, mechanisms, and frameworks to guide policymaking, set system parameters, and shape decisions on mobilizing, allocating, and managing LTC resources.	Mobilizing, directing, and managing monetary and non-monetary resources to support the effective delivery of LTC services and support. This involves strategies for revenue generation, pooling and resource allocation, and expenditure management, including establishing financing arrangements that combine private and public sources.	Training, recruiting and retaining, and deploying personnel to deliver LTC services and support. This includes defining LTC-specific roles and competencies, providing targeted training, developing effective workforce strategies, and aligning supply with care demand.	Planning, delivering, and managing a full continuum of care. This covers health and social care services delivered in homes, communities, residential facilities, and hospitals by public and private providers; non-service support such as cash benefits, assistive devices, home modifications, and related training; and informal caregiver support; among other.

Source: World Bank (forthcoming).

Section III: How is climate change affecting people with LTC needs?

This section begins by presenting the Intergovernmental Panel on Climate Change’s hazard, exposure, vulnerability, and response framework and its relevance to LTC systems. It then describes the main climate hazards affecting the ECA region, followed by a discussion of the sources of vulnerability to climate hazards specific to older people and people living with disabilities, as a means of exploring the vulnerability

⁸ World Bank. “The ABCs of Long-Term Care: A Functional Framework for Building System Capacities in Europe and Central Asia. Forthcoming.

of people with care needs to climate hazards. The section concludes with an elaboration on the individual health and care impacts of climate hazards on these populations and the implications for LTC systems.

Section IIIa: Hazard, exposure, vulnerability, and response framework

The IPCC climate risk framework articulates the interplay of hazard, exposure, vulnerability, and response to assess the potential adverse impacts of climate change. This framework first defines three risk determinants: hazard, exposure, and vulnerability. Hazard refers to physical climate risks that may have adverse effects (e.g., heatwaves, floods, prolonged droughts, and rising temperatures); exposure refers to the presence of people, assets, or systems in an area in which climate hazard events may occur;⁹ and vulnerability is the propensity of exposed populations or systems to harm when impacted by climate hazard events, determined by factors such as health status, socioeconomic conditions, and infrastructure quality¹⁰ (Cardona et al. 2012). The inclusion of response, as the fourth risk determinant, in the most recent version of the IPCC framework expands it to account for the actions taken to adapt to or mitigate the effects of climate hazards, including emergency preparedness, adaptive service delivery, and long-term adaption planning, making this version particularly relevant for a consideration of the role of LTC systems in promoting the climate adaptation of people with care needs (IPCC 2023).

As illustrated in Figure 1, the combination of these four risk determinants—hazard, exposure, vulnerability, and response—influences the extent to which people are adversely impacted by climate hazards. Differences in these determinants mean that some populations are more adversely impacted than others. The risk of being adversely impacted depends not only on the exposure or vulnerability of a population or the effectiveness of a response, but also on the frequency, intensity, and severity of the climate hazard. These hazards include climate-related extreme weather events, such as heatwaves or wildfires, and slow onset events, such as gradually increasing temperatures or prolonged droughts. They influence exposure and vulnerability in different ways, often requiring different responses.

Figure 1: Interaction between determinants of climate risk



Source: Simpson et al. (2021).

⁹ For example, such as older adults living or LTC facilities located in a landslide-prone area or coastal region where flooding occurs.

¹⁰ For example, persons with mobility-related disabilities, who may encounter difficulties evacuating without assistance during flooding. See the section below for a fuller discussion of the sources of vulnerability to climate hazards.

This framework provides the basis to better understand the risk of climate change to people with care needs and LTC systems. Disruptions to LTC provision arise from the interplay between climate hazards (e.g., the frequency and intensity of heatwaves or flooding), exposure (e.g., LTC facilities located in flood-prone areas), vulnerability (e.g., reliance on uninterrupted electricity and water supplies), and response (e.g., lack of protocols for extreme heat or inadequate infrastructure adaptation to extreme heat). In this context, the type of climate hazard will also shape the interplay between exposure, vulnerability, and response. While extreme weather events may cause immediate disruption to LTC systems, slow onset events may cause cumulative strain over time—gradually weakening infrastructure, increasing operational demands, or degrading service quality. Where these risk determinants converge, the adverse impacts to LTC systems increase and progressively imperil those who rely on LTC services and support (Figure 2). These risk factors similarly converge for individuals, creating new types of care needs and potentially greater demand for LTC.

The remainder of this section is organized around the IPCC framework. It first reviews climate hazards across ECA countries, followed by an assessment of the vulnerability of older people and persons with disabilities. Subsequent sections examine the dimension of response—both within LTC systems and through coordination with other sectors.

Section IIIb: Climate hazards in ECA

Europe¹¹ is the fastest-warming continent worldwide, resulting in adverse impacts across subregions.¹²

The warmest year on record for Europe was 2024, with above-average temperatures across most of the continent and the most extreme heat and drought conditions in the southeastern areas. In July 2024, Southeastern Europe recorded its longest heatwave on record, lasting 13 consecutive days, with an average temperature anomaly of 9.7° Celsius (C) and affecting 55 percent of the area. The frequency and intensity of extreme heatwaves are projected to increase. In a global warming scenario of 1.5°C, extreme heat could cause up to 30,000 deaths annually in Europe, with Southeastern Europe projected to experience the highest and fastest-growing number of heat-related deaths (C3S and WMO 2025; Bednar-Friedl et al. 2022).

Europe is projected to have one of the largest increases in flood risk and extreme precipitation, even as parts of its subregions experience increasing droughts and water scarcity. Projections indicate an increase in the frequency and intensity of lower-than-average river flows, exacerbating risks to soil moisture, droughts, and growing challenges in water supply. Under a 1.5°C warming scenario, Southeastern Europe is expected to see a slight rise in the number of days with water scarcity and drought,

¹¹ This section considers two broad sub-regions in ECA. In this analysis, data referring to “Europe” are primarily drawn from the *European State of the Climate 2024* (ESOTC), produced by the Copernicus Climate Change Service (C3S) and the World Meteorological Organization (WMO). The ESOTC covers two overlapping regions: the C3S domain (focused on the EU) and the broader WMO domain, which includes the Western Balkans, the South Caucasus (Armenia, Azerbaijan, Georgia), and parts of the Middle East. The data used from the ESOTC 2024 in this paper include both the Western Balkans and the South Caucasus. For more information on the specific subregions see: <https://climate.copernicus.eu/esotc/2024/about-report>

¹² This is not an exhaustive discussion of climate hazards in ECA, focusing rather on those climate hazards for which there is evidence of their impacts on older people and people with disabilities, as discussed in the sections below. For a fuller consideration of climate change in ECA, see [Climate Change in Europe and Central Asia](#).

potentially exposing 18 percent of its population to at least moderate water scarcity. Similarly, extreme precipitation and surface flooding are projected to increase in all regions of Europe. In 2024, Storm Boris brought exceptionally heavy rainfall and triggered one of the most damaging widespread floods on record in Europe, impacting eight countries across Central and Eastern Europe—including Croatia, Romania, and Poland. These eight countries experienced the equivalent of up to three months of rainfall within four days (C3S and WMO 2025; Bednar-Friedl et al. 2022).

Central Asia¹³ is facing growing climate risks, including rising temperatures, changing rainfall patterns, and more frequent heatwaves and droughts, with water scarcity among the region’s most critical climate-exacerbated challenges. Rising temperatures are heightening the risk of heatwaves, aggravating droughts conditions in arid and semiarid areas of Central Asia. In contrast, the mean precipitation levels remain almost unchanged. Over the past 50 to 60 years, Central Asia has seen a 30 percent reduction in glacier surface area due to climate change. This glacier retreat has intensified the frequency and intensity of climate-related hazards such as floods and landslides. However, the most pressing challenge lies in increasing water scarcity, exacerbated by rising temperatures and shrinking water sources, with Turkmenistan and Uzbekistan being the most water-stressed countries in the subregion. Given that a significant portion of Central Asia’s population lives in drought-prone areas, water scarcity poses a widespread risk—contributing to food and water shortages and rising rates of malnutrition. Projections under global warming scenarios of 1.5°C, 2°C, and 3°C indicate a growing number of people will become increasingly vulnerable to water-related impacts, including reduced freshwater availability, degradation of water quality, and associated risks to food security, nutrition, and human health (ADB 2022; Shaw et al. 2022).

¹³ In this analysis, “Central Asia” refers to the group of five countries defined as such in Chapter 10 (Asia) of the IPCC Sixth Assessment Report: Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan. The data used are primarily drawn from this source. See page 1463 of the report for more details:

https://www.ipcc.ch/report/ar6/wg2/downloads/report/IPCC_AR6_WGII_Chapter10.pdf

Box 3: Summary of key climate hazards in ECA

EUROPE

- **Extreme heat:** Fastest-warming continent with heatwaves becoming more frequent and intense. Southeastern Europe is projected to experience the highest and fastest growing number of heat-related deaths.
- **Floods:** More frequent and intense heavy rainfall and flooding across all regions (e.g., Storm Boris in 2024).
- **Droughts and water scarcity:** Increasing drought risks and growing challenges in water supply. Up to 18 percent of Southeastern Europe's population is at risk of water scarcity under 1.5C warming.

CENTRAL ASIA

- **Droughts and water scarcity:** Most pressing challenge, especially in Turkmenistan and Uzbekistan, with cascading risks to food security, nutrition, and health.
- **Extreme heat:** Rising temperatures and more frequent heatwaves.
- **Glacier retreat:** Around 30 percent of surface loss in the past 50 to 60 years, driving floods and landslides.

Source: Authors, based on listed sources (2025).

Section IIIc: Vulnerability to climate hazards

Some populations are more vulnerable than others and thus experience greater impacts upon exposure to climate hazards, as illustrated above in Figure 1. Vulnerability may arise from biological and health, social, and systematic and structural factors. Biological and health-related factors, such as physiological changes, pre-existing health conditions, and reduced functional ability, increase susceptibility to harm (Prina et al. 2024). Social factors, including living alone and limited social support networks, can undermine the ability of a person to cope when a climate hazard occurs. Systemic and structural factors, such as poverty, outmigration, poor infrastructure, political instability, exclusion from emergency planning, and inadequate access to information, compound risks (Prina et al. 2024; Kosanic et al. 2022).¹⁴ Together, these factors can make people with care needs more vulnerable to the impacts of climate hazards than the general population. This poses unique challenges for countries in ECA, as they face rising care needs due to rapidly aging populations.

Older people and people living with disabilities are particularly vulnerable to climate hazards due to biological, health, and social factors. For example, older people often have difficulty regulating their body temperature,¹⁵ limiting their capacity to adapt to extreme heat or cold. This means that older people are at greater risk of experiencing adverse health impacts compared to the general population during climate-related extreme weather events and slow onset events. Older people and people living with disabilities, including those with dementia and intellectual disabilities, may underestimate or not recognize the dangers of climate hazards. As a result, they may not take appropriate protective actions, which may

¹⁴ These can also undermine response efforts, as discussed in the sections below.

¹⁵ See discussion in the next section below.

exacerbate the impact of climate hazards (Ratwatte et al. 2022; Prina et al. 2024). Social factors, such as living alone or lacking supportive networks, also amplify vulnerability. For instance, during the 2003 European heatwave, a significant number of older adults died alone in their homes, many of whom were unaware of the risks posed by the prolonged period of hotter weather or were unable to seek help (Hajat et al. 2010).

Systemic and structural factors may also hinder the ability of these two populations to effectively cope with or adapt to climate change, including through their exclusion from response planning. Socioeconomic challenges, such as poverty and limited financial resources, reduce individuals' access to essential services and support and to quality housing, heightening their vulnerability to climate risk. Inadequate infrastructure, including limited public transportation and poorly constructed or maintained housing,¹⁶ compounds these vulnerabilities by restricting mobility and reducing access to essential services and support during crises. Additionally, limited availability and quality of LTC provision may leave many older people and people with disabilities without reliable assistance during climate events. Older people often live in areas with limited access to cooling centers, flood defenses, and emergency evacuation routes, making them particularly susceptible during heatwaves, floods, and other extreme weather events (Prina et al. 2024). Outmigration of younger, more active individuals from a region can leave people with care needs increasingly isolated and with reduced social support. These factors may undermine disaster response efforts. Studies have found that older people were disproportionately affected during heatwaves and floods in part because emergency systems failed to identify and reach those living alone, with mobility challenges, and/or without digital access (UN 2021). A review of nearly 1,700 articles on climate adaptation responses reported that only 1 percent considered the needs of people living with disabilities (Global Disability Inclusion Report 2025).

Section IIId: Direct impacts of climate hazards on health and functional ability

Extreme variations in temperature, hot or cold, can have negative implications for well-being among people with LTC needs. Challenges in body temperature regulation are common among people with care needs: older people have impaired thermoregulatory control, while people with disabilities experience similar challenges, arising from medication or underlying medical conditions.^{17,18} Excessive heat exposure can lead to heat-related illnesses, such as heatstroke, heat exhaustion, heat cramps, heat syncope, heat edema, and heat rash. Heat exposure can result in severe dehydration, acute cerebrovascular accidents,

¹⁶ During the 1995 Kobe (Japan) earthquake, individuals aged over 60 experienced higher mortality rates; this was due to them residing in collapse-prone wooden homes with heavy roof tiles and sleeping on the ground floor (Pollock & Wartman, 2020; Kunii et al., 1995).

¹⁷ Very young children also have difficulty regulating their body temperatures.

¹⁸ The human body relies on a thermoregulatory system to monitor and respond to thermal stress, ensuring thermal comfort (Heal & Park, 2016). When exposed to extreme temperatures, the body adapts through physiological responses such as vasodilation or vasoconstriction, which regulate heat loss or retention. While not all health conditions are directly influenced by temperature fluctuations, research indicates that sudden temperature changes can significantly affect morbidity and mortality in certain diseases (Sugg et al., 2016). These regulatory mechanisms make circulatory, respiratory, metabolic, and neoplastic diseases particularly susceptible to temperature-related health risks (White, 2017). Additionally, certain diuretics, beta-blockers, and antipsychotic drugs increase the risk of dehydration, hypotension, and heat stroke (Kenny et al., 2010).

and contribute to thrombogenesis; it can also further aggravate chronic pulmonary conditions, cardiac conditions, kidney disorders, and psychiatric illness (WHO 2015a). Critically, in extreme cases, these impacts can lead to mortality. Without climate adaptation measures, Honda et al. (2013) estimate that annual heat-related deaths in Central Asia will increase by 139 percent by the 2030s and 301 percent by the 2050s. Exposure to cold can lead to increased risk of hypothermia and frostbite.

These direct health impacts frequently translate into exacerbated functional limitations. For instance, severe heatstroke can lead to lasting neurological damage, impacting mobility, cognition, and self-care abilities. Similarly, the aggravation of chronic conditions like cardiac or respiratory illness due to extreme temperatures can impact cognitive function among older adults, leading to increased dependency. Consequently, effectively addressing these climate-induced health challenges necessitates good LTC provision—one that includes strong coordination with acute health provision, given that individuals often transition between acute care settings for immediate health crises and LTC for ongoing support and rehabilitation, as set out in Figure 2.

Climate hazards can also contribute to the increased transmission and outbreak of infectious diseases. Droughts and floods, for example, can lead to water-borne illnesses such as cholera, E. coli infections, and dysentery (Stanke et al. 2013), which can have severe impacts on people with care needs. Insect-borne diseases, such as malaria and encephalitis, can see increased transmission during drought events and are more likely to develop in older people and people with pre-existing conditions (Bogovič et al. 2018; Del Prete et al. 2019; Hansen 2020).

Such climate-driven infections have profound consequences for groups with care needs, especially older people, initiating a complex cascade of health deterioration. For one, these infections can accelerate the aging process itself, potentially through increased inflammation and damage to tissues (Gavazzi and Krause 2002). This impact hits individuals who have experienced declines across many bodily systems, leaving them with a reduced ability to cope with stress like a severe illness (Fried et al. 2001). This often means they can no longer manage daily tasks on their own, leading directly to a need for intensive LTC support.

Air pollution can result in a higher frequency and severity of respiratory illnesses. During wildfire events, smoke inhalation can cause or exacerbate respiratory illness among older people and people living with disabilities (Liu et al. 2016; Wollschlaeger et al. 2022). Hospitalization rates can also increase significantly for these two populations following exposure to wildfire smoke, for both individuals near wildfire outbreaks as well as those living thousands of kilometers away from the outbreak (McBrien et al. 2023; Le et al. 2014). Beyond the acute impacts of such events, long-term exposure to air pollution is associated with increased functional limitations and a faster decline in physical functioning related to aging (De Zwart et al. 2018).

The mental health of older people and people living with disabilities can be affected by climate hazards, particularly during evacuations or disruptions to care services. During evacuations due to climate-related extreme weather events, older people and people living with disabilities may experience heightened stress and negative mental health outcomes, which can have long-lasting effects. Older people who experienced traumatic stress, such as a difficult evacuation during Hurricane Sandy in the United States, were more likely to have limited functioning six years later (Pruchno et al. 2019). It is also possible for climate-induced

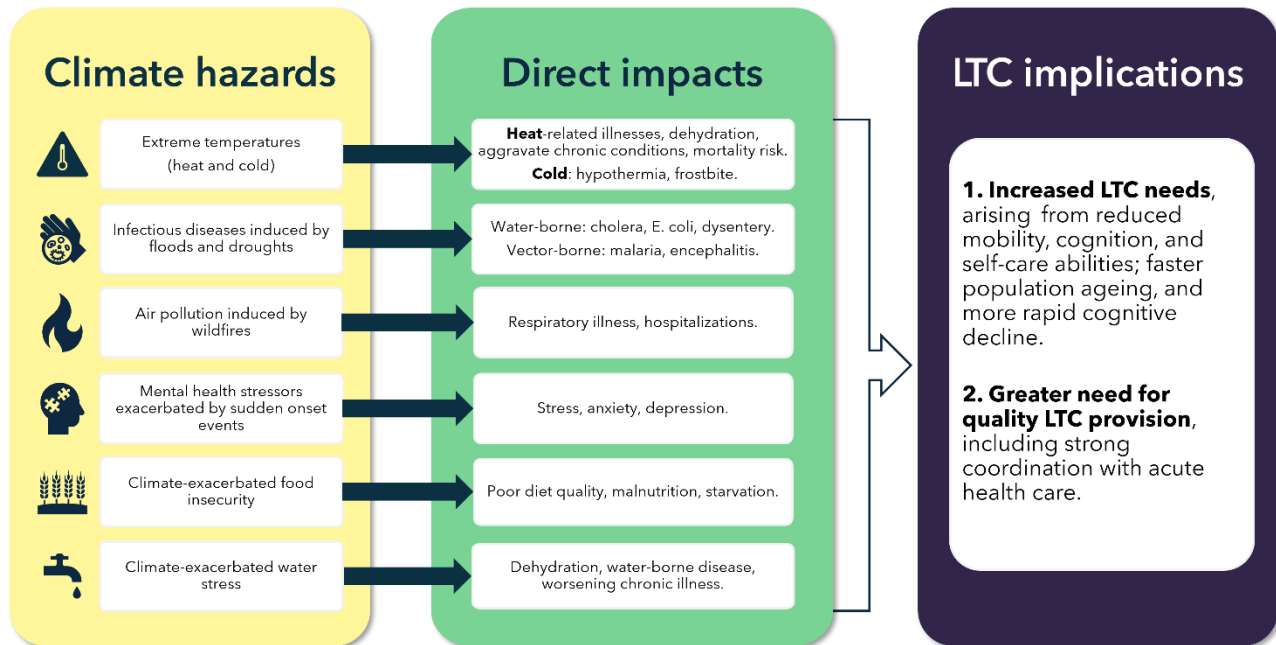
stress and displacement to accelerate cognitive decline among older people and people living with disabilities (Hu et al. 2024; Mitchell et al. 2024). These consequences disproportionately affect the mental health of older people (Ayalon et al. 2021).

Climate-induced food insecurity may significantly increase the risk of malnutrition and starvation, particularly for older people and people living with disabilities. When climate hazards disrupt food production and distribution, people may struggle to access food supplies, which can lead to malnutrition and even starvation. Under conditions of disrupted food production, food prices typically rise (Hallegatte et al. 2016), which presents challenges for older people and people living with disabilities living on fixed incomes. Low-income older adults with functional limitations are more likely to report poor diet quality, due to difficulties in accessing and preparing food (Nazri et al. 2021). The impact of climate-exacerbated food insecurity on older people and people living with disabilities is further compounded by socioeconomic factors such as lower income, limited education, social isolation, and poor health conditions (Elia and Stratton 2005). From an LTC perspective, malnutrition is notably associated with a decline in general functional status and decreased bone mass, immune dysfunction, delayed recovery post-surgery, and high hospitalization and readmission rates, among other (Favaro-Moreira et al. 2016).

Water stress intensified by climate hazards can result in significant health risks for older individuals and people with disabilities. Water stress can be understood as a situation where there is not enough water of sufficient quality to meet the demands of people and the environment. Water stress can lead to dehydration, transmission of water-borne disease, malnutrition, worsening of chronic conditions, and worsened mental health (WHO 2023), with suggestion of increased mortality rates among older people.¹⁹ Notably, dehydration can lead to acute issues like confusion, delirium, and increased fall risk, while also being associated with long-term cognitive decline, including the progression of dementia (GuideStar Eldercare 2024). When climate hazards occur, water stress can be exacerbated.

¹⁹ Berman et al. (2017) conducted a retrospective study on medical data from 618 counties in the U.S. (2000–2013) and found a 1.55% increase in mortality among individuals over 65 during severe and worsening drought conditions. Salvador et al. (2021) studied mortality trends in Lisbon (1983–2016) and found that drought periods were associated with heightened mortality rates due to natural, respiratory, and circulatory causes, with the oldest populations experiencing the highest risk.

Figure 2: Direct impacts of climate hazards on health and functional ability, and their implications for LTC systems

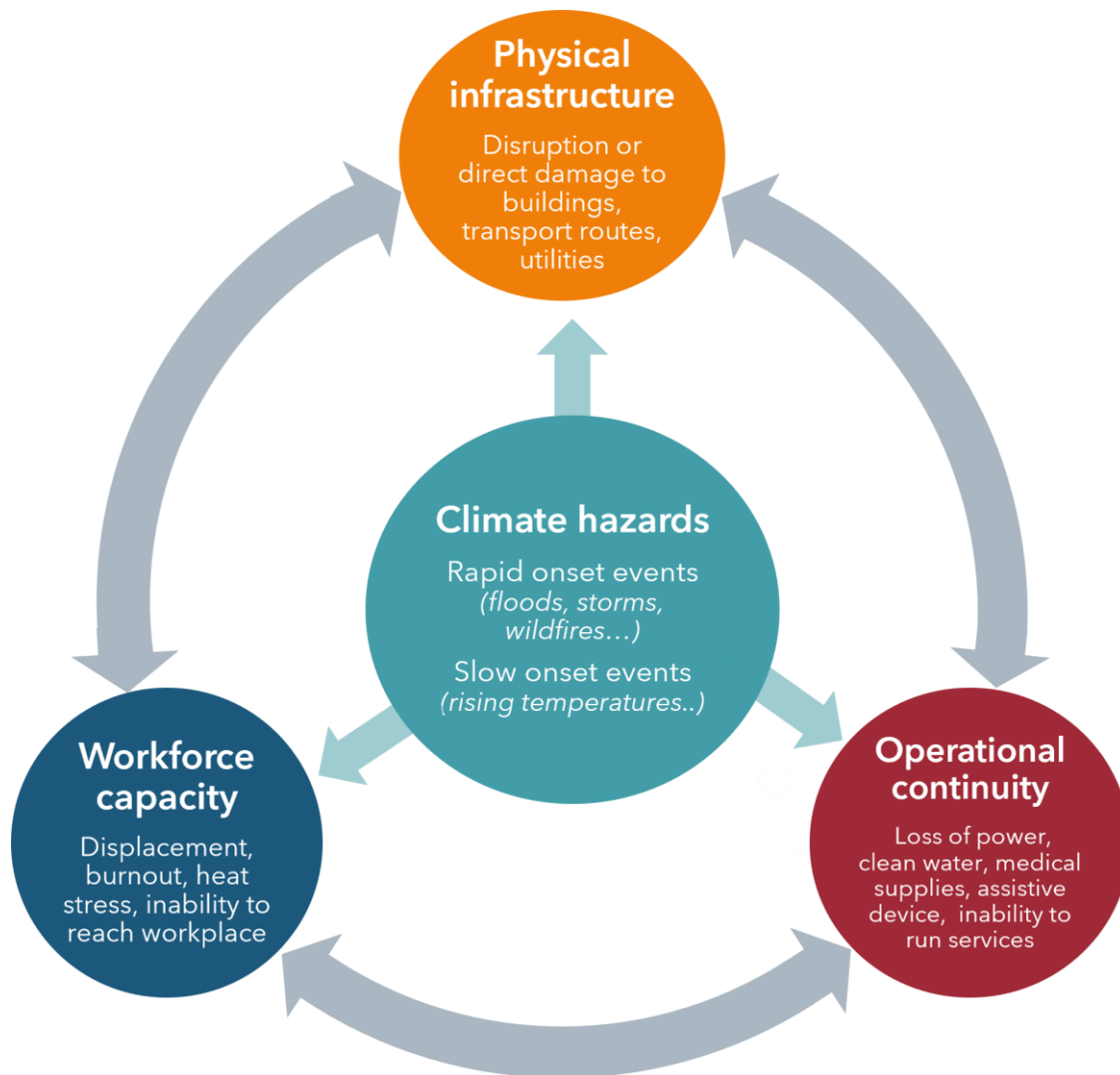


Source: Authors (2025).

Section IIIe: LTC system impacts

Climate hazards pose profound challenges to LTC systems, impacting them across three critical and interconnected dimensions: physical infrastructure, operational continuity, and workforce capacity. Physical infrastructure encompasses tangible assets like facilities, utility networks, and transportation routes, all of which face progressive deterioration or direct damage and disruption from climate hazards, rendering care environments unsafe or inaccessible. Operational continuity refers to the sustained delivery of essential care services, which is severely disrupted by the loss of vital resources like power, clean water, food, or medical supplies, even if physical structures remain intact. Meanwhile, workforce capacity, crucial for care delivery, is compromised when personnel cannot reach facilities due to hazardous conditions, face personal displacement, or experience heightened burnout. These elements are interdependent: infrastructure damage can directly halt operations; compromised operational environments can impact the ability of staff to work safely and effectively; and a depleted workforce can limit service delivery. Consequently, climate impacts on any one dimension create cascading vulnerabilities across the entire LTC system, compromising its ability to deliver consistent and quality care (Figure 3).

Figure 3: Climate hazard impacts on LTC systems



Source: Authors (2025).

Beyond structural damage, climate hazards can disrupt the operational functionality of LTC facilities and home- and community- based care, thereby hindering the provision of care services and support. Rapid onset events, such as flooding or wildfires, can damage and destroy LTC facilities (Madani Hosseini et al. 2024a). Storms and high winds, among other climate hazards, may damage the electricity grids supplying power to LTC facilities and private homes where LTC services are delivered.²⁰ As a result, the beneficiaries of LTC services may experience service disruptions in the absence of a power backup system. For example, during Winter Storm Uri (2021) in the U.S. state of Texas, 10 percent of nursing homes and 33 percent of assisted living facilities lost power. Such power outages and disruptions in water supply were associated with higher mortality rates among residents in nursing homes (Downer et al. 2025). Following Hurricane

²⁰ In rare cases, the provision of cash benefits may also be disrupted.

Irma in 2017, the U.S. state of Florida enacted legislation requiring nursing homes and assisted living facilities to have generators that could maintain the supply of electricity for multiple days. Similarly, slow onset events, such as progressively rising temperatures, can gradually strain infrastructure and reduce the availability of basic resources. Disruptions or reduced access to clean water, as may be the case when home care services are provided in rural communities experiencing water stress, can further comprise the ability to provide essential care. Disrupted care access for individuals who rely on medical equipment, such as ventilators and dialysis machines, in LTC facilities or at home may also occur (National Council on Disability 2023). These disruptions may comprise essential care, including access to medications and assistive devices, resulting in significant risks to health and safety.

Climate hazards can also result in displacement of formal and informal caregivers, undermining their capacity to deliver care services and support to these vulnerable groups. When climate hazards force evacuations or disrupt transportation networks, people receiving LTC may lose access to formal and informal caregivers. Professional staff working in LTC facilities, professional home care providers, and informal caregivers are all at risk of extreme heat, particularly for those working in poorly ventilated spaces, wearing personal protective equipment, or spending time outdoors. In France, for example, home helpers are listed among the occupations most exposed to extreme temperature (Benhamou and Flamand 2023). Heat stress can cause physical injuries, among other health effects. In the absence of access to a qualified LTC workforce, beneficiaries of LTC services and support may be forced to rely on other or “ad-hoc” informal caregivers that may lack the specialized training and experience to appropriately support these populations or risk losing access to needed care.

Section IV: LTC systems responses to the impact of climate hazards

Climate hazards have pressed a number of OECD countries to develop interventions to protect people with LTC care needs, with direct relevance to LTC systems. The following section presents case studies from the United Kingdom, Spain, France, Japan, and the United States. These case studies do not reflect a systematic review of climate interventions focused on people with care needs nor the complete LTC systems in these countries. Instead, the examples aim to illustrate practical strategies that could be adapted or piloted in ECA countries to promote climate adaptation through formal LTC systems or to reach informal caregivers. These are largely descriptive as evidence of their effectiveness is thin. As discussed above, these countries were chosen because their LTC systems are well documented and because they represent some diversity in terms of policies response, institutional arrangements for and financing mechanisms of LTC. They also present strategies for responding to a range of climate hazards; while extreme heat is a pressing concern in all countries, cold weather, flooding, climate disasters, and air quality are also considered. In section 5, these strategies are then situated within a broader review of the literature to identify a set of interventions to promote climate adaptation through LTC systems in ECA countries.

Section IVa: United Kingdom

The United Kingdom Health Security Agency (UKHSA) has developed extensive resources to prepare for and respond to the risk of heatwaves and cold spells. The centerpiece of UKHSA’s response is the Weather-Health Alerting System,²¹ which comprises both Heat-Health Alerting (HHA) and Cold-Health Alerting (CHA) functions. The Weather-Health Alerting System aims to provide early warning notifications to the health and social care sectors, the community responding to crises and disasters, the voluntary sector, and government agencies.²² Among other critical resources, this alerting system, specifically the HHA and CHA, provides guidance documents, referred to as “action cards,” for health and social care workers attending to people receiving LTC services and support.²³ These action cards feature preparedness activities that should be carried out all year-round, preparedness activities for extreme heat and cold periods—which correspond with summer and winter, and actions to take under different weather severity levels (Table 2)

Table 2: Heat-Health Alerting System levels and impacts	
Levels	Impact
Green (Preparedness)	A Green Alert indicates that weather impacts are likely to have minimal impact on health.
Yellow (Response)	A Yellow Alert indicates that weather impacts are unlikely to affect most people but could impact those who are particularly vulnerable.
Amber (Enhanced Response)	An Amber Alert indicates that weather impacts may affect the whole population.
Red (Emergency Response)	A Red Alert indicates that weather impacts are likely to be of significant risk to life for even the healthy population.

Source: United Kingdom Health Security Agency (2024).

One key goal of the impact-based approach is to enable health and social care professionals and managers within these systems to make informed decisions. Professionals and managers in health and social care sectors are alerted to take action to reduce impacts on people receiving LTC services and support earlier than the general population (UKHSA 2024a). One evaluation of the HHA found that while the system appears to work well as an important planning and alerting tool for emergency planners and health and social care providers, “alert fatigue” can limit effectiveness among professionals (Roberts et al. 2022). In other words, people begin to stop paying attention to messaging if alerts are sent too frequently. This suggests that maintaining the effectiveness of such alert systems requires balancing the frequency and clarity of communications to ensure they remain actionable and impactful.

²¹ The WHA system is managed by UKSHA in partnership with the Met Office, the UK’s national meteorological service.
²² The Heat-Health Alert (HHA) operates from 1 June to 30 September and the Cold-Health Alert (CHA) operates from 1 November to 30 March. In addition to the Weather-Health Alerting System providing early warning information to the organizations listed above, the public and other organizations may register for alerts from these systems.
²³ Action Cards also provide guidance to professionals who work with older people and people with disabilities outside the LTC system, and public health officials, who are encouraged to identify populations vulnerable to heat, for example.

Box 4 presents extreme heat long-term preparedness measures for health and social care professionals and LTC facility managers offered by UKHSA. Table B4.1 characterizes the impact-related information provided to health care and social care professionals due to hot weather.

Box 4: UKHSA Heat-Health Alerting long-term preparedness

Complementing the Heat-Health Alerting System, UKHSA offers a series of specific measures that health and social care professionals and LTC managers can take to ensure preparedness for heat emergencies (UKHSA 2025), as described in the table below.

Table B4.1: UKHSA heat-health alerting long-term preparedness measures

Measure	Description
Identify those at risk	Establish methods to identify, alert, and monitor individuals most at risk of heat-related illnesses on your caseload (whether for clinical or social reasons). Consider seeking clinical advice if you are unsure whether someone is at risk.
Develop business continuity and hot weather plans	Work with partner agencies, commissioners, providers, and businesses to coordinate business continuity plans and hot weather plans, ensuring those most affected by heat-related illness are appropriately supported. This includes developing and maintaining a register of the most vulnerable individuals to ensure they can be offered appropriate support where required.
Update care records	Develop approaches to include in care records factors that may exacerbate vulnerability to heat and consider whether changes to care plans might be necessary in the event of hot weather (e.g., initiating daily visits by formal or informal care givers for those living alone). Ensure plans are in place to be able to monitor room temperature, and, in clinical settings, body temperature, pulse rate, blood pressure, and hydration levels.
Training plans	Develop a training plan to ensure that staff are aware of the local hot weather plans and prepared to take actions that are appropriate for a given alert level.
Workplace champions	Consider identifying individuals to be workplace champions to increase staff awareness of the potential risks of adverse weather and climate change and to encourage staff to embed best practice mitigation against, planning for, and responding to adverse weather.
Data-sharing agreements	Ensure appropriate data-sharing agreements or protocols are in place to allow organizations to share data or intelligence, including details of those most vulnerable to heat-related illness where appropriate. For example, for the care sector, having these arrangements in place with local authorities could be particularly important.
Surveillance and monitoring	Consider surveillance and monitoring to evaluate the effectiveness of any hot weather interventions that are established locally, including locally relevant indicators and metrics.
Environmental improvements	Consider environmental improvements that need to be made to provide a safe environment during hot weather, through a facility risk assessment. For example, checking that windows can be opened safely and shaded, preferably by curtains with pale, reflective lining rather than by metal venetian blinds and curtains with dark linings, which can make conditions worse. Consider outside shading, as this can aid cooling.

Source: United Kingdom Health Security Agency (2024a).

UKHSA also provides guidance to help people stay warm during the winter and highlights how older people and people living with disabilities are particularly vulnerable to cold weather. UKHSA offers a targeted cold weather action card for managers working in the social care sector, namely managers of

registered care homes and other residential settings and managers of home care providers (UKHSA 2024b). This guide is described in Table 3. In addition to this guidance to formal care providers, UKHSA describes interventions that any person could adopt to protect themselves and the people around them from cold weather, such as maintaining a warm home, seeking financial support, cold-proofing homes, and checking on their neighbors. This guidance recognizes the vulnerabilities of various populations, including older people, and is disseminated through government websites and social media, local governments, nongovernmental organizations, and the media.

Table 3: UKHSA cold weather response guidance for social care sector managers	
Setting	Risk reduction measure
LTC facility, home	Develop, and where necessary implement, business continuity plans (BCP) for cold weather, and ensure all staff know how to take action accordingly.
LTC facility, home	Take simple measures to protect clients and staff from the cold, snow, and/or ice.
Home	Identify who is at higher risk of cold-related illnesses and how to reduce that risk.
Home	Have a plan in place for individuals who are responsible for keeping the client warm during the winter, and ensure staff know how to raise concerns regarding clients whom they are working with.
Home	Ensure staff are aware of the main sources of support for housing, energy bills, and other needs to signpost clients to, including support for those on low incomes.

Source: United Kingdom Health Security Agency (2024b).

UKHSA also offers, among other, guidance for health and social care providers to prepare for, respond to, and recover from flooding. UKHSA’s flood guidance recognizes that older people and people with pre-existing health conditions may be especially vulnerable during and after flooding (UKHSA 2025). The guidance notes that flood risks may include immediate risks, such as drowning, trauma, and infections from contaminated water, while long-term effects may include respiratory illnesses due to mold growth and psychological distress. The guidance also details actions that health and social care providers can take before, during, and after flooding (UKHSA 2025). Flood preparedness includes ensuring that LTC facilities have structural risk assessment and response plans in place, monitoring flood alerts and warnings, protecting vulnerable populations, securing essential supplies, and communicating flood safety advice. During flooding, actions comprise emergency plan activation, evacuation support, mental health support, and health monitoring. In flood recovery settings, UKHSA recommends post-flood health checks, mold and environmental health risk assessments, support for displaced individuals, and long-term mental health interventions.

Section IVb: Spain

In response to the significant mortalities experienced during the 2003 summer heatwaves, Spain implemented a comprehensive plan to reduce the impact of extreme heat. The National Plan of Preventive Actions Against the Effects of Extreme Temperatures on Health (National Plan), adopted in 2004, is annually revised and activated between May and September. It acknowledges that vulnerable

populations, including older adults, people with disabilities, and those experiencing social marginalization, isolation, dependency, or inadequate housing conditions, face heightened risks that compound their exposure and reduce their ability to cope with the impacts of extreme temperature. To address this, the plan includes a specific care program aimed at identifying and protecting those at greatest risk, with a strong focus on older people—particularly during extreme heat events, through coordinated social services and tailored interventions.

The National Plan outlines a strategy for preventing, controlling, and managing risks associated with extreme heat through coordinated institutional responses. Implementation of the National Plan is led by the Ministry of Health in close cooperation with other ministries, including, the Ministry of Social Rights, Consumer Affairs, and the 2030 Agenda and the Institute for the Elderly and Social Services (IMERSO).²⁴ The plan is structured around levels of risks resulting from rising temperatures, which are monitored daily by the national meteorological agency (Agencia Estatal de Meteorología, AEMET).

The LTC system in Spain is largely decentralized; thus, a key aspect of the National Plan is coordination of services and support across the central, regional, and local authorities. The National Plan operates through the LTC system by disseminating information to health and social care professionals to increase awareness and understanding of the risks of extreme heat and to provide guidance on appropriate response. The LTC system plays an important role in identifying and providing care to populations with care needs during extreme heat events. Networks of municipal services identify populations with care needs and manage home care, telecare, and social centers, among others. During extreme heat events, the LTC system disseminates actionable information to the municipal networks of health and social care professionals and coordinates across services providers.

Within the Spanish LTC system, local authorities develop interventions tailored to local populations. In this context, the Community of Madrid is responsible for developing and implementing an action plan for social health centers, including LTC facilities, to prevent and address the impacts of extreme heat on people with care needs. The Community of Madrid's Specific Intervention Plan for Socio-Health Centers (Intervention Plan) includes a list of preventive measures for long-term planning and improvement of infrastructure and facilities year-round, in preparation for the summer season, specifically aimed at the directors and staff of LTC services. The Intervention Plan is structured across three alert levels (green, yellow, and red) and proposes a comprehensive list of measures to be undertaken at each level (Table 4). Heat risk levels are determined by the national meteorological agency (Agencia Estatal de Meteorología, AEMET). Table 5 describes the response measures, with associated descriptions, for LTC facilities when the Red Alert is activated.

²⁴ The Institute for the Elderly and Social Services (IMERSO) is an agency within the Spanish Ministry of Education, Social Politics and Sports. IMERSO's responsibilities include policy development, research, international cooperation, information dissemination, management and monitoring of certain public social services for older people and people living with disabilities (Government of Spain 2025)

Table 4: Community of Madrid heat risk levels and impacts

Risk level	Impacts
Green (Normal ranges)	Temperatures are predicted not to exceed 36.5° Celsius (C). No increase to health risk is expected, and routine operations continue with standard precautions.
Yellow (Caution)	Temperatures are predicted to exceed 36.5°C for at least three days. Moderate heat effects are possible, particularly for populations with care needs.
Red (High risk)	Temperatures are predicted to exceed 38.5°C for one day or 36.5°C for at least three days. Under these conditions, serious heat risks are possible for populations with care needs.

Source: Community of Madrid (2024).

Table 5: Community of Madrid Intervention plan response measures in LTC facilities: Red Alert

Ensure facilities remain as cool as possible	Ensure the measures for Levels 1 (Green) and 2 (Yellow) have been properly implemented.
	Increase the shading outdoors; watering the soil can help cool the surrounding air.
	Keep the curtains and windows closed when the outside temperature is higher than the inside temperature.
	When the outside temperature is lower than the inside temperature, open the windows; this may not happen until late at night or in the early hours of the morning.
	Recommend avoiding physical activity and outdoor outings during the hottest hours of the day (from 11:00 am to 3:00 pm).
	Regularly monitor indoor temperatures in all areas where patients reside during the hottest periods.
	Ensure staff are available to assist and advise residents.
	Utilize cooler nighttime temperatures to ventilate and cool the building.
	Lower indoor temperatures by turning off unnecessary lights and electrical devices.
Monitor health status and conditions of residents	Consider adjusting visiting hours to mornings or late afternoons/evenings to avoid the afternoon heat and overcrowding.
	Monitor body temperature, heart and respiratory rate, blood pressure, and hydration levels.
	Watch for signs of headache, unusual fatigue, weakness, and dizziness.
Reduce heat-related health risks	Keep an eye out for any changes in behavior, especially increased drowsiness.
	Encourage residents to stay in the cooler areas of the facilities as much as possible.
	Move residents so that each one spends time in the cool room/area (below 26°C), prioritizing and allocating additional time to high-risk residents or those showing signs of distress (including increased body temperature); for those who cannot be moved or who might become too disoriented by being moved, take measures to cool them down (e.g., fluids and cold compresses) and increase monitoring.
	Regularly monitor residents' fluid intake, ensuring they receive cold drinks, particularly if they require assistance to drink. Emphasize the importance of increased fluid consumption during high-temperature periods to mitigate the risk of bloodstream infections from Gram-negative bacteria. For individuals on high doses of diuretics, oral rehydration salts are advisable. Additionally, bananas, orange juice, and occasional salty snacks can aid in replenishing salts lost through sweating.

	Advise residents to avoid caffeine (e.g., coffee, tea, and cola), overly sweet beverages, and alcohol.
	Encourage residents to wear light, loose-fitting cotton clothing, which helps absorb sweat and prevents skin irritation.
	Frequently apply cold water to exposed areas of the body, either by spraying or wetting them. Additionally, placing a damp cloth on the back of the neck can assist in regulating body temperature.
	If possible, offer access to cold showers or baths.

Source: Community of Madrid (2024).

Section IVc: France

High mortality rates during the 2003 European heatwave spurred the French government to put in place a National Heatwave Plan. The 2003 heatwave resulted in 15,000 deaths in France, 82 percent of which were among individuals 75 years or older. Within a year, the French government developed and implemented the National Heatwave Plan. It is based on three complementary pillars: (i) raise awareness among the general population and professionals, including those working in health care and social services, through communication and training; (ii) reduce heat exposure for the most vulnerable; and (iii) take immediate preventive action during the most extreme episodes (Government of France 2014; Pascal et al. 2018). As a result, the country experienced a 90 percent reduction in excess mortality²⁵ between 2003 and 2019 (Ford 2019).

The National Heatwave Plan includes mandatory and voluntary protocols, including for LTC facilities and home care providers. Those called upon to take preventive action include health and social care professionals and regional and local officials. Preventive measures include implementing emergency plans in LTC facilities and ensuring vulnerable populations at home can be reached and offered support when needed. The plan includes both recommended and mandated measures (see Table 6) to be taken by actors within the French LTC system (Pascal et al. 2010). Among the mandated measures, local governments must create databases of vulnerable people to ensure contact with them when temperatures rise. While the creation of the database is mandatory, registration is voluntary; vulnerable individuals, their families, and caregivers are encouraged to register by the municipality, health care providers, social workers, and nongovernmental organizations. The registries typically record a person's name, address, contact details, and information on their vulnerabilities (i.e., age, health conditions, and living situation). This information is used only to contact the person when a response is triggered.

²⁵ According to the WHO, excess mortality is defined as the difference between the total number of deaths estimated for a specific place and given time period and the number that would have been expected in the absence of a crisis.

Table 6: Legal requirements related to long-term care in France's National Heatwave Plan

Legal citation	Legally required measure
Decree No. 2005-768 of July 7, 2005	All institutions hosting older people or people living with disabilities must define the organization, role, and responsibilities of the institution during a heat wave.
Decree No. 2005-768 of July 7, 2005	All institutions hosting older people or people living with disabilities must at least have access to a cool room.
Law No. 2004-626 of June 30, 2004, Decree No. 2004-926 of September 1, 2004	Each city must create a database of vulnerable populations who should be contacted in a timely manner by social services during a heat wave; being on the list is a voluntary action. Vulnerable people include people over the age of 65 and people living with disabilities; individuals register voluntarily.

Source: Pascal et al. (2010).

Preventive actions are then triggered, as needed, based on the Heatwave and Health Alert System.

France activates its Heatwave and Health Alert System (SACS) annually from June 1 to September 15. On June 1, all local government units are put on the first alert level (Green Alert).²⁶ This triggers preparatory activities, and the national meteorology agency (Météo-France) starts producing daily weather alert maps. When an Amber or Red Alert is triggered, Public Health France carries out rapid health surveillance to identify any unusual impacts to inform the ongoing response. Notably, these higher alert thresholds trigger actions by LTC facilities—including the mobilization of additional human resources, and by local governments—including outreach to vulnerable people, facilitated by the information in the registry. Given that many older people in France live alone, such support may include calls or visits from volunteers—including students or municipal workers – during Amber or Red Alerts, who can connect them with a doctor.

A new national plan was presented in 2023 based on lessons learned in the interim. This new plan, which was prepared by the French Ministry for Ecological Transition and Territorial Cohesion, is drafted around four axes: limiting the impact on the daily lives of French citizens; ensuring the continuity of essential public services; ensuring the continuity of economic life; and protecting natural environments and resources. It stresses attention to dissemination of prevention messages on public transportation and via SMS during periods of high temperatures. In addition, special measures are envisioned to encourage people who act as legal guardians to ensure that individuals under legal protection are registered in the municipal databases of vulnerable populations to be supported during heatwaves, as described above. Additional information is distributed among health and LTC professionals to enable the identification of vulnerable people,²⁷ who are defined as:

- Old age, dependency or disability, social isolation, precariousness, social exclusion;
- Lack knowledge about preventive measures, are inappropriately clothed, consume alcohol;

²⁶ There are four levels: green, yellow, orange, red.

²⁷ https://www.ecologie.gouv.fr/sites/default/files/documents/08.06.2023_DP-PlanChaleur.pdf

- Presence of certain chronic conditions and use of certain medications that may interfere with the body's ability to adapt to heat; history of heat-related disorders; and,
- Presence of memory disorders, mental disorders, behavioral disorders, difficulties with understanding and orientation, among others.

The new plan directs those with LTC needs to, among others, limit temperature increases in the home; avoid exposure to heat; cool down regularly, stay hydrated, and adapt their diet; and avoid medications that are particularly damaging to the kidneys for dehydrated patients.

Section IVd: Japan

In recent years, the Japanese government has taken measures to better focus on older people and people living with disabilities in responding to climate hazards. The measures include implementing a heat warning system, updating disaster response planning to focus on older people and people living with disabilities, and encouraging local peer-to-peer support networks—an innovative supplement to traditional post-disaster support. The country's heightened agility to target local interventions for climate hazards is a result of its standalone LTC system working in close partnership with local municipal authorities. The discussion here focuses on older people who receive services through Japan's LTC system as well as older people who do not, but who are recognized as being particularly vulnerable during disasters.

In 2021, the Japanese Ministry of Environment introduced the Heat Stroke Alert (HSA), a national early warning system that draws public attention to heat dangers and encourages individuals to take actions to prevent heat illness. The HSA provides specific guidance for older people and people living with disabilities (Ministry of Environment, Government of Japan 2025), which is available on dedicated websites. Table 7 outlines HSA measures that illustrate how the needs of older people and people with disabilities are a focus for the Japanese government in responding to climate hazards.

Table 7: Japan's heat stroke alert - heat illness measures	
Population	Measures
General	Avoid going out as much as possible and keep the room cool with an air conditioner.
General	Avoid exercising as a general rule.
General	Check the heat index in your immediate surroundings and take action to prevent heat illness.
Older people, general	Encourage the elderly around you to use air conditioners and the like, regardless of whether it is daytime or nighttime.
Older people, people living with disabilities, other vulnerable groups	Older people, children, people with chronic illnesses, obese people, and people with disabilities are more susceptible to heat illness. Taking breaks and staying hydrated can help reduce the impact of extreme heat.

Source: Ministry of Environment, Government of Japan (2025).

The Japanese government has updated disaster planning and response mechanisms with respect to people with care needs and older people more generally. The Disaster Countermeasures Basic Act, which

underpins Japan's disaster risk management system, and associated legislation, explicitly requires preparation and response measures to protect vulnerable groups, including people with care needs. These measures have evolved, based on lessons learned from major disasters (Okuda and Tomio 2024). Currently, municipalities are required to identify and register people who may face difficulties during evacuations (World Bank 2022). These registries, which include contact information, the reasons why additional support is needed, and, more recently, individual evacuation plans, are used to support evacuations and/or confirm the safety of people. Prior to a disaster, preparatory measures must include plans to ensure the safety and well-being of people with care needs in shelters or evacuation centers. This includes providing the necessary care services and supplies as well as establishing protocols to coordinate with people who remain at home. To facilitate coordination between the disaster risk management units and the long-term care system, some municipalities have started to use the opportunity of people applying for long-term care insurance to prepare individual evacuation plans alongside the preparation of individual care plans (Okuda and Tomio 2024).

Japan has also taken steps to strengthen the climate resilience of the LTC systems. Following devastations caused by a tsunami and an earthquake in 2011, the Government of Japan adopted measures which include provisions to strengthen information communications, emergency power generation and alternative water sources, and the storage of water, food, and other necessities in healthcare and welfare institutions, including LTC facilities, in the event of a disaster, including those caused by climate change. Business continuity plans, which must be put in place prior to any disaster, together with regular training and drills, aim to minimize disruptions to LTC services and support. These are now part of the accreditation criteria for nursing care facilities and recommended for other elderly care facilities (World Bank 2024). Local Japanese government agencies also have responded to the impacts of climate hazards on older people by supporting peer-to-peer support systems, a unique post-disaster service. Under this approach, local authorities enable older people to visit peers living in temporary (post-climate disaster) housing to offer social support and mitigate isolation (Yotsui et al. 2016).

Section IVe: United States

In the United States, state governments proactively plan for and respond to the risk of climate change for people with LTC needs. The United States Environmental Protection Agency, a federal government agency, maintains resources suggesting measures that can be adopted to protect older people from climate hazards. Suggested responses are directed at older people, who may or may not be receiving formal LTC services, via a dedicated website. U.S. state governments take proactive approaches to address the impact of climate hazards on older people and people living with disabilities. The Commonwealth of Massachusetts Department of Public Health (MDPH), for example, has developed Climate Hazard Adaptation Profiles (CHAPs) that can be used to support local authorities in understanding and responding to climate-related health impacts (MDPH 2025). There are five CHAPs: Extreme Heat and Poor Air Quality, Sea Level Rise, Inland Flooding, Extreme Weather, and Recreational Air Quality. Each CHAP highlights populations that are most vulnerable, including older people and older people who are residing alone, and people living with disabilities, and offers intervention measures. Table 8 presents an example of climate

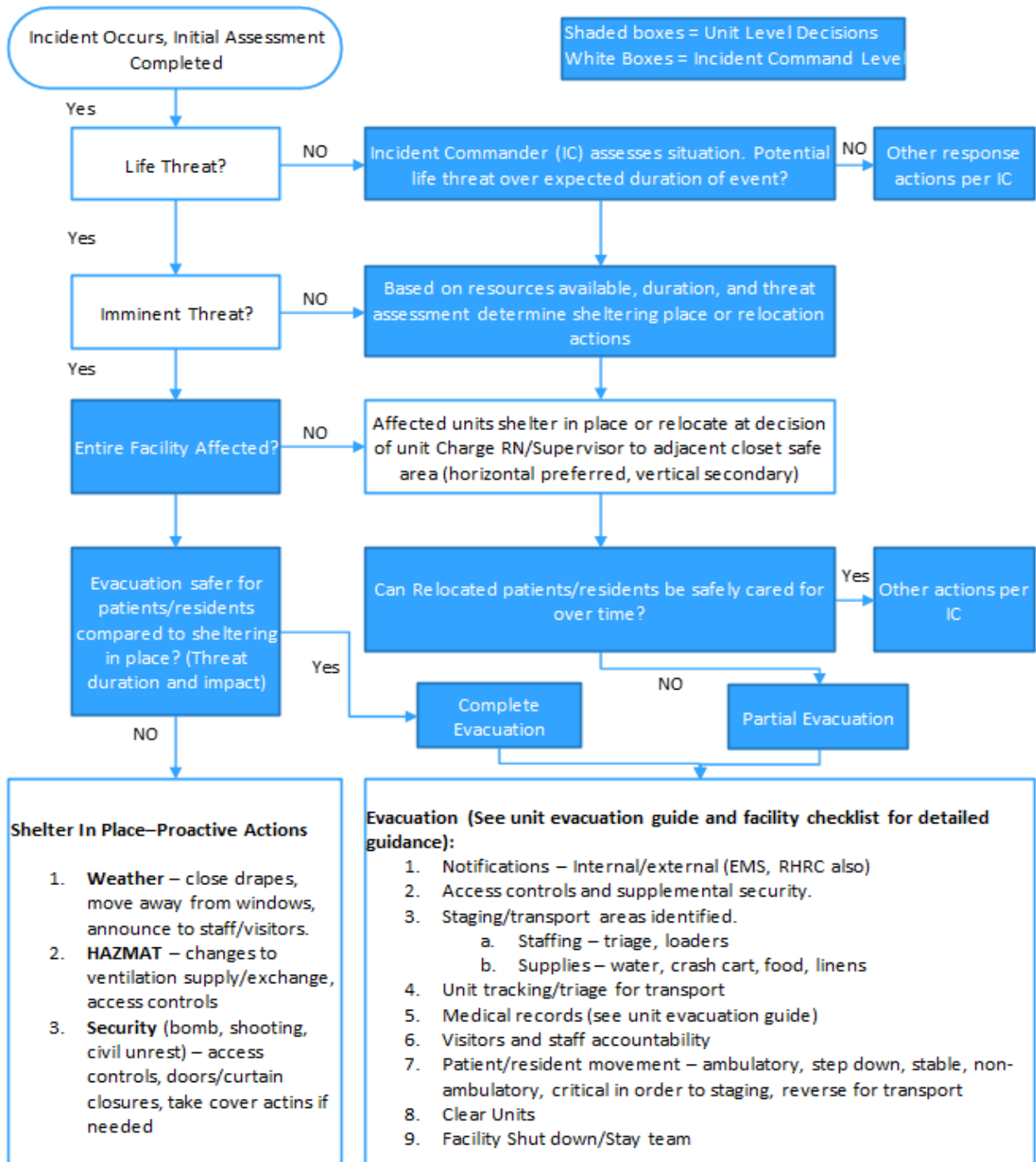
hazard intervention strategies offered by MDPH, drawing on the Extreme Heat and Poor Air Quality Events CHAP.

Table 8: Extreme Heat and Poor Air Quality Events in the Climate Hazard Adaptation Profiles of the Massachusetts Department of Public Health (United States) LTC-related Measure
Identify vulnerable populations and health issues in your community using the DPH Community Profiles , and other tools available on the EPHT website .
Identify and map vulnerable locations and populations using DPH's Climate Change Vulnerability Mapping Tool .
Determine if cooling centers can operate during loss of electricity.
Advertise availability of cooling centers.
Provide transportation to cooling centers for vulnerable residents.
Improve public access to air conditioning units and develop other cooling strategies to reduce exposure to heat.
Raise awareness of heat and air pollution-related health risks among medical care providers and residents by monitoring local air quality conditions using MassDEP's MassAIR Online and USEPA's AirNow .

Source: Massachusetts Department of Public Health, United States (2025).

The state of Minnesota illustrates how state governments can offer resources to address the impact of climate change on LTC facilities. The Minnesota Long Term Care Preparedness Toolkit (Preparedness Toolkit) provides sample templates, forms, and suggested resources to develop and/or enhance the emergency preparedness plans of LTC facilities and agencies throughout the state (Minnesota Department of Health 2023); it focuses on people who receive LTC services and support. Figure 4 is the Preparedness Toolkit and provides an example of how systematic decisions can be made at LTC facilities when climate hazard events occur.

Figure 4: Sheltering, relocation, and evacuation decision tree



Source: Minnesota Department of Health, United States (2023).

Section IVf: Learning from selected OECD interventions

The case studies above draw attention to examples of how LTC systems are taking action to promote climate adaptation among people with care needs. The case studies reviewed above, while not exhaustive, illustrate examples of governments modifying their LTC systems in recognition of how a range of climate shocks and stressors are or have the potential to negatively affect people with care needs. These examples point to the central concern of many OECD countries with the deadly effects of extreme heat for older people and people with disabilities, as well as the impacts of cold weather, flooding, and air quality (Table 9). These measures are informed by the evidence of how climate change is affecting older people and people with disabilities, most vividly through national disasters which promoted action, as was the case in France and Spain. Many of these are relatively recent adaptations and, thus, not yet rigorously evaluated; although there is suggestion of their effectiveness (Box 5). Frequently, the case studies present measures that were introduced through formal LTC services, complemented by efforts to extend the same knowledge and information to informal care providers and people with care needs enabling them to act—and often to all older people and people with disabilities, recognizing their unique vulnerabilities to climate hazards. The relative weight of these strategies tends to reflect the organization of the LTC system itself, with a strong focus on formal LTC providers in the United Kingdom, Spain, France, and Japan and more emphasis on empowering people with care needs and their families to protect themselves in the United States.

Frequently, LTC services and support are adapted to buffer the impact of climate change on people's care needs. Many of the case studies describe new provisions that countries have introduced into their existing LTC services and support. These are most consistently described in terms of protecting people with care needs from rising temperatures, which include measures to identify people vulnerable to heat, alert them to increasing temperatures, monitor their health and well-being, and take action to keep them cool and/or address any arising health issues. Adjustments to LTC facilities are also common, as is the case in France, which requires LTC facilities to maintain heatwave responses plans and access to cooling rooms. Box 5 offers evidence of the effectiveness of the response to extreme heat in selected EU member states. Contingency plans are increasingly a feature of LTC facilities and home care providers to reduce the impact of extreme climate events on their services, such as the examples of preparing for flooding in the United Kingdom and the United States. Examples of new, innovative services also emerged: In Japan, locally supported peer-to-peer post-disaster support systems seek to meet a climate hazard-related need for older people, while France has mandated cities to create and maintain registries of vulnerable people to help rapidly identify and reach people with appropriate support.

The provision of guidance and training for frontline staff, informal care providers, and individuals is a central feature of these initiatives. Given the focus on introducing modifications into existing LTC services, it is unsurprising that most case studies include the creation of guidance notes, supported by training, for frontline staff, given the requirement that they use new sources of information, such as weather data and alert maps, communicate new guidance to their beneficiaries, and/or carry-out new tasks. The UKHSA's guidance documents, which target frontline health and social care workers to help them identify, monitor, and respond to climate-related risks among older people and people with disabilities, is one such example. These efforts support workforce preparedness and capacity-building, which are essential for delivering

effective care during climate-related events. The resource from the state of Minnesota is similarly aimed at strengthening the capacity of care professionals to make decisions that protect LTC facility residents.

In all cases, strong governance enables action by service providers, and, critically, promotes cross-sectoral collaboration with new ministries or agencies. Strong leadership, at the central or local level, translates into national policy commitments and local implementation plans. Establishing a policy commitment to responding to climate change through LTC systems or protecting people with care needs from climate change lays the foundation for action. This is the case in France, where the national heat wave plan sets clear policy direction first for vulnerable populations, including older people and people with disabilities, and then the population more generally. The National Plan and Fundamental Plan, in Spain and Japan respectively, provide the basis for collaboration between levels of government to protect people from extreme heat in Spain and for disaster response in Japan. Cross-sectoral coordination with meteorological agencies is common, given the need to ensure a regular stream of reliable weather information to inform action by service providers and care staff. In the UK, the Weather-Health Alerting System reflects coordinated national leadership, interagency collaboration, and integration of climate risk into public health planning.

Table 9: Selected OECD country climate hazard interventions		
OECD country/sub-region	Measure	Climate hazard
United Kingdom	Extreme weather alerting system	Extreme heat/ heatwaves/ extreme cold
United Kingdom	Heatwave preparedness	Extreme heat
United Kingdom	Flood preparedness and response	Flooding
Japan	Heat alerting system	Extreme heat/ heatwaves
Japan	Peer-to-peer support networks	Sudden-onset events
France	National heatwave response plan	Extreme heat/ heatwaves
Spain	National heatwave response plan	Extreme heat/ heatwaves
United States: Federal	Localized action and information sharing	General climate impact reduction guidance
United States: Commonwealth of Massachusetts	State-wide and localized action and coordination; information sharing	Extreme heat, air quality, sea level rise, flooding, extreme weather, recreational air quality
United States: State of Minnesota	Sheltering, relocation, and evacuation decision tree	Emergency planning guidance

Source: Authors (2025).

Box 5: Evidence from Europe points to the benefits of harnessing LTC systems for climate response

As evidenced by a recent World Bank review, policy responses in EU countries help to reduce heatwave-related mortality (World Bank 2025). To varying degrees and subject to local conditions, heatwave responses in France, Spain, Italy, Germany, and Switzerland each rely on LTC institutions and workers to plan, disseminate information, directly respond, and provide post-heatwave support to people with care needs, many of whom are particularly vulnerable to the extreme heat. As a result, France's national heatwave plan helped to reduce mortality in the 2006 heatwave compared to 2003 by 68 percent, after accounting for temperature differences. Implementation of Spain's national heatwave plan with regional adaptations contributed to a 10-15 percent reduction in daily mortality during heat events compared to pre-intervention years. Heatwave responses in Italy, Germany, and Switzerland also saved lives. In Italy, a national heat-warning system and city-specific alerts and measures helped to reduce heat-related mortality in major cities by 9-15 percent, while Germany's heat health response systems contributed to reductions in emergency admissions and excess deaths during alert-triggered heatwaves. Switzerland's national and cantonal efforts enabled a 27 percent reduction in excess mortality among older people during the 2015 heatwave compared to prior relevant years.

Source: World Bank. Unlivable: Extreme Heat in Cities in Europe and Central Asia, forthcoming.

Section V: Adapting LTC systems to climate change

In response to the increasing prevalence, frequency, and intensity of climate hazards, governments in the ECA region have options to reduce the impacts of climate hazard on people with care needs. This section considers adaptations to LTC systems that can help minimize disruption to care and respond to emerging care needs among LTC beneficiaries arising from climate change. It also considers strategies that LTC systems could adopt to reach people affected by climate events who are outside the formal LTC system and for whom the LTC system may be an appropriate response. The strategies are structured into three categories: before, during (i.e., immediately prior to and while the event is occurring), and after a climate hazard event (Figure 5).²⁸ Building on the case studies and a broader review of the literature, these strategies and interventions span across the functions of an LTC system, recognizing that LTC systems in ECA at varying stages of maturity and contending with climate hazards of varying frequency and intensity.

Section Va: Before climate hazard events

Identify who may need care support during climate hazard events. Among LTC providers, establishing methods to identify and monitor beneficiaries most at risk of climate events will help ensure that they receive services and support quickly when needed. Such steps can be formally integrated into individual care plans by developing protocols to include factors that may exacerbate vulnerability to climate hazards, such as heat or cold, in care records. This is the case of the UK. Given the continued reliance on informal care arrangements and family members in many ECA countries, maintaining registries of all older people and people with disabilities in localities exposed to climate events can help ensure that information and services reach a broad range of people quickly and appropriately. In France, for example, three categories of people living at home may register in local heatwave vulnerability databases: persons aged 65 and older;

²⁸ While the “before, during and after” structure points to preparing for and responding to discrete extreme climate events, the strategies discussed are also relevant for responding to slow onset events.

persons over 60 years of age recognized as unfit for work; and disabled adults, professionals, or persons close to them (Government of France 2024).

Establish care protocols to respond to emerging LTC needs that arise due to climate hazards. Climate change requires revisiting how existing services and support are delivered, ensuring they are adapted to protect the health and well-being of people with care needs. Protocols may specify actions for different climate hazards and severity levels, tailored to each LTC setting, and should be embedded within minimum service requirements or quality standards. In the preparation stage, for a potential hot or cold spell, for example, care providers may be required to monitor the temperature in a person's home and advise on or take actions to help ensure that a room remains cool, such as helping older people to hang blinds or open windows, or on how to remain warm. At a higher level of severity, care providers may be required to monitor the body temperature, pulse rate, and blood pressure of their patients to identify signs of health stress and then take steps to cool the person down, as described in the case of Spain when a Red Alert is activated. In addition, home care providers can conduct assessments of home environments and where possible, refer LTC users to programs to improve climate resilience in private homes (Young and Bergseng 2020) or their energy efficiency. Peer-to-peer networks could also be developed in advance to provide physical and emotional support during climate hazard events, such as in the case of Japan.

Set up weather alerting systems so that appropriate responses to climate hazards via LTC systems are quickly triggered. Tiered weather alerting systems, which are tied to specific guidance to LTC staff and managers, enable LTC systems to rapidly modify care services in response to climate hazards, thereby buffering the impact on beneficiaries. Ensuring this alerting system reaches informal care providers and people with care needs expand its reach. The examples from the United Kingdom, Spain, France, and Japan present alerting systems for extreme heat, and from the UK for extreme cold. Similar alerting systems could be created for flooding or severe water stress that leads to droughts, as has been done for social assistance systems elsewhere in the world.²⁹ In all cases, such alerting systems require close collaboration between the meteorological agencies and ministries responsible for LTC.

Establish appropriate ways of communicating with people with care needs in residential facilities, in the community, or at home. Given that LTC services and support are provided across a range of settings, clear messages need to be established in advance and communicated using mechanisms that effectively reach all people with care needs. Messages on preventative actions in the face of climate hazards are central to the five country cases. Key communications should reach people receiving formal LTC services via their caregivers. This means that formal caregivers should be early and consistent recipients of climate hazard response messaging and that LTC providers are a critical component in information dissemination for people outside of the formal LTC system and the general public. Communication pathways can include traditional and social media, SMS messaging, and public transit announcements; targeted messaging via the legal representatives of people with care needs and those registered in local databases can help extend the reach. In addition, a dedicated telephone hotline can assist vulnerable populations to obtain more information on climate hazard risks and responses.

²⁹ See: Bowen, T. et al. (2020).

Build informal and formal caregiver capacity to reduce the impact of climate hazards on people receiving care at home. Equipping informal and formal caregivers in home settings with knowledge and training on climate hazards and impact reduction measures is key (C40 Cities 2022). This is particularly important, as home care workers have access to homes which no one else may regularly enter and thus may be the sole means of identifying, alerting, and monitoring people with care needs vulnerable to climate hazards. The measures to be adopted by care providers could be integrated into the curriculum for pre- and in-service training and built into the standards for home care provision. Opportunities to reach informal providers could include messaging, as discussed above, supplemented by information provided through any formal training or support provided to caregivers by the LTC system.

Ensure quality service provision and continuity for people in LTC facilities by strengthening caregiver capacity. LTC managers and workers in LTC facilities should have opportunities to increase their knowledge and skills with respect to how climate hazards impact people with care needs and when and how to introduce service modifications to protect people from these impacts. These opportunities could be integrated into the curriculum for pre- and in-service training for LTC staff and built into the quality standards for LTC facilities, which would then be monitored and assessed. Good emergency planning and training for LTC facility staff is also necessary, including for sheltering in place, evacuation, and communication with families, as demonstrated in the case of flooding in the United Kingdom and the United States.

Climate-resilient LTC facilities need to be established in advance and offer services during climate hazard events. If LTC facilities are required to maintain emergency backup power sources, they can continue delivering services, even when local power grids are impacted. Governments can help LTC facilities improve their resilience to heat-related and wildfire climate hazards by requiring passive and active cooling measures, fire-retardant building materials, and enhanced ventilation. In preparation for flood and heavy rain-related climate hazards, measures such as the use of waterproof building materials, elevated electrical and drainage systems, and backup filtration measures can strengthen resilience to climate hazards. These measures need to be integrated into the country's quality assurance framework, which sets the standards for infrastructure (including location), staffing, and equipment, like the approach taken in Japan to incentivize the adoption of business continuity plans. Canada's emerging LTC climate adaptation standards recommend resilience features such as backup energy systems, refuge rooms, fire-retardant materials, and elevated electrical systems in flood-prone areas. These measures are supported by government and academic guidance that encourage a shift toward long-term sustainability and safety in care facilities (Wollschlaeger et al. 2022).

Integrate information technology into LTC system intervention measures to reduce the impact of climate hazards. Internet access and the rapid adoption of artificial intelligence (AI) present opportunities for governments to work closely with providers to expand the use of IT solutions in LTC system responses to climate hazards. For example, remote electronic activities of daily living monitoring can help older people and people living with disabilities to experience a heightened sense of security (Gutman et al. 2010). Encouraging widespread implementation of electronic monitoring solutions could enable LTC systems to efficiently target resources to those who need them most during climate hazard events. In addition, integrated AI-operated thermal comfort systems that combine integrated sensors (e.g., wearable and/or

infrared), infrastructure for enabling system interoperability, learning and control algorithms, and actuators (e.g., HVAC system set points and ceiling fans) (Ghahramani et al. 2020) may be an emerging pathway for LTC systems to respond to increases in average temperatures, as well as heat waves.

Ensure close coordination between LTC systems and relevant public and private sector actors. The five countries in the case studies exemplified strong government leadership in establishing collaboration between the LTC system, the national meteorological agency, and government ministries responsible for disaster risk management, emergency response, and/or climate adaptation. By explicitly highlighting in national climate resilience and adaptation plans how climate hazards impact people with care needs (e.g., by considering the needs of older people and people living with disabilities) and the role LTC systems play in responding to these needs, governments can create incentives for productive collaboration across ministries. A strategic basis for cross-sectoral coordination of climate hazard responses can help break down barriers and encourage effective planning and implementation across sectors. Formal governance structures that include LTC systems and other key actors (e.g., coordinating committees) could be developed to institutionalize such collaboration to prepare for and then respond to climate events.

Section Vb: During climate hazards events

Ensure continuity of care by focusing on people with care needs during climate-related evacuations and emergencies. Emergency response plans should include targeted evacuation routes and communication tools, disability-inclusive emergency shelters, and measures for those reliant on assistive technology or regular medical treatment to keep receiving care (Simpson et al. 2021). Backup power generators for LTC facilities and shelters should be activated and emergency supplies accessed. Human resources capacity should be expanded by activating formal and informal caregiver networks, where these exist, to reach and support people with care needs and, potentially other older people and people living with disabilities who may now need care because of the climate hazard. Mobile health units and telemedicine solutions are emerging strategies to maintain service continuity in climate-vulnerable areas (Gotanda et al. 2015).

Support formal and informal caregivers in applying targeted interventions, given individual needs. During heatwaves, for example, caregivers should help older people with body cooling, by applying ice towels, minimizing clothing or wearing cotton clothing saturated with water. Fans should only be used with application to the body at air temperatures greater than 38°C because older people tend to sweat at lower rates (WHO 2021). Where possible, older people and people living with disabilities should use air conditioners or access cooling centers and be supported in actively remaining hydrated.

Disseminate climate hazard emergency information through trusted local leaders and activate local support networks, including organizations of persons with disabilities. During climate hazard emergencies, people with functional limitations and their caregivers must make quick decisions. As choices in such stressful situations are influenced by trusted people, including faith leaders, social workers, case managers, medical professionals, and frontline professionals, governments would be well placed to rely on these trusted voices to transmit climate risk communications (LTSS Center 2024).

Coordinate with disaster risk management, emergency response, and climate change agencies. Actions taken by the LTC system to protect existing beneficiaries from climate hazards or to extend support to

additional people should be closely coordinated with the government ministries or agencies responsible for disaster risk management and emergency response. As discussed above, such coordination mechanisms should be established well in advance of any climate event and be underpinned by national legislation or policy that sets out roles and responsibilities, communication channels, and coordination mechanisms. In this way, the response through the LTC systems contributes toward a broader multi-sectoral response, while recognizing the unique vulnerabilities of people with care needs.

Section Vc: After climate hazard events

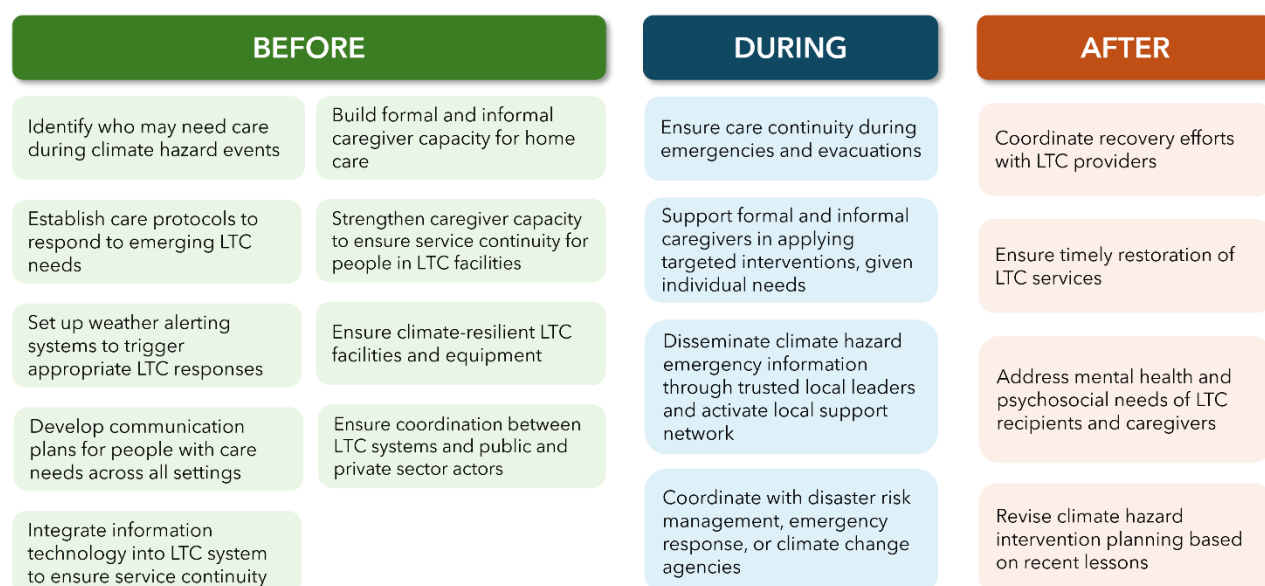
Coordinate recovery efforts with LTC providers. By ensuring that post-climate hazard recovery planning explicitly includes LTC providers, such as home care agencies, LTC facilities, formal caregivers, and informal caregiver networks, governments can help people with functional limitations receive timely support as services are restored and living conditions stabilize. This requires coordination across sectors, as well as setting out a role for LTC providers in recovery efforts.

Focus on the restoration of LTC services. Post-climate hazard recovery plans should encompass the resumption of LTC services and support, including the reopening of LTC facilities, home visits, and rehabilitation services. If the existing LTC service infrastructure is damaged, temporary care solutions should be provided. However, people with care needs should not be unnecessarily institutionalized and/or segregated from the general population. Where possible, people with functional limitations should be supported in returning and remaining in their homes and communities.

Provide for the mental health and psychosocial support needs of people with LTC needs and their caregivers. Climate hazard events can be traumatic for those affected. By offering targeted services to address psychological and social challenges, such as losing one's home and local support networks, LTC systems can help address these often overlooked, but critical needs in post-climate hazard recovery phases.

Update climate hazard intervention planning, given the lessons learned from recent experiences. Regular assessments to determine what went well and what could go better when the next climate-related event occurs is an important practice, informed by established monitoring and evaluation, as possible. Input into future climate hazard intervention planning should ideally come from all relevant stakeholders, including formal and informal caregivers, LTC system managers, and other involved government officials, as well as beneficiaries of LTC services and support, and other older people and people with disabilities. Lessons learned should shape existing planning efforts and be implemented, where feasible. Beyond informing future responses, such analysis should inform the design and delivery of regular LTC services and support, quality standards and staffing, among other aspects of the LTC system, to ensure that the system continues to adapt to the changing climate.

Figure 5: Measures and interventions to strengthen LTC systems before, during, and after climate hazard events



Source: Authors (2025).

Section VI: What is the scope for future work?

Consideration of climate change within LTC systems is an emerging field, including for policymakers in the ECA region. While this working paper aims to identify strategies and interventions that LTC systems in ECA can adopt to promote the climate resilience of the populations they support, the analysis faced limitations. First, the working paper does not include a systematic scoping of existing interventions in the five OECD countries or in other countries. Second, this working paper considers evidence of the impact on and vulnerability of older people and people with disabilities to elucidate the ways in which people with care needs may be affected by climate change. This approach, while providing the basis for an evidence-based discussion, may miss impacts or vulnerabilities that are unique to people with functional limitations, which are not well documented in the literature focused on these two populations. Recognizing these limitations, based on the analysis in the sections above, future research could consider the following questions:

- **How effective have climate hazard responses through LTC systems in other countries been?** Comprehensive evaluations of interventions designed and implemented by LTC systems in other countries would help guide policy for climate hazard response in ECA countries. By examining what measures are effective, are still experimental, and/or should be further analyzed, policymakers in ECA would have more relevant guidance for adapting the interventions discussed above to ECA-specific conditions and populations, as well as a better understanding of implementation challenges and potential trade-offs.
- **How have informal caregivers in ECA responded to the impact of climate hazards?** Informal caregivers are typically the critical service providers in most LTC systems. A systematic examination and tabulation of best practices of how informal caregivers protect and support people with

functional limitations could be a valuable input into LTC system climate hazard response planning and implementation such input would provide evidence and potentially innovations upon which to build the design and delivery of interventions to strengthen informal caregiver capacity to reduce the impact of climate hazards on people receiving care at home.

- **How does institutional coordination between LTC and emergency response influence effectiveness?** While recognizing that LTC systems may be standalone or coordinated through the social protection and health systems, examination of how the LTC system coordinates with the emergency response system (or not) could provide lessons on how to best manage the response to extreme climate events for people with functional limitations. Such analysis would inform the design of care protocols for responding to emerging LTC needs that arise due to climate emergencies and strategies to promote coordination between the LTC system and emergency response system.
- **How do climate hazards generally affect people with care needs in ECA?** While the impacts of climate hazards on older people and people living with disabilities are generally understood, these may be unique for people with functional limitations. There may also be local, regional, and/or group characteristics that moderate or exacerbate climate impacts across ECA countries. Knowledge of these differences would help to design appropriate intervention strategies for LTC systems in the ECA region, as could estimates of the number of people with care needs vulnerable to different climate hazards across countries in ECA.
- **Apply an adaptive social protection (ASP) framework to strengthen analysis.** Consideration of the role that LTC systems can play in promoting climate adaptation among beneficiaries, their families, and older people and people with disabilities more broadly suggests that these systems are part of the ASP landscape in a country. From this perspective, future analysis could apply the World Bank's ASP framework and its four pillars: (i) programs and delivery systems, (ii) data and information, (iii) financing, and (iv) institutional arrangements (Bowen et al. 2020). This would build out the analysis presented in this working paper, promoting a greater focus on data and information systems, financing, and institutional arrangements, which span those for LTC (i.e., social protection and health), disaster risk management, and emergency response.

Section VII: Conclusion

Climate change is already affecting people with functional limitations, and LTC systems offer a natural platform to promote climate adaptation. The evidence and country examples reviewed in this paper underscore three central messages. First, older people and those with LTC needs are disproportionately affected by climate hazards, including heat and cold, floods, fires, air pollution, and water scarcity; and these impacts exacerbate functional limitations. Second, climate hazards threaten LTC service delivery, through impacts on infrastructure, operations, and workforce capacity. Third, strengthening climate resilience does not require building entirely new services; it is primarily about adapting existing ones, guided by timely climate risk information and coordinated action across sectors.

Integrating climate adaptation into ongoing LTC reforms offers a double dividend: systems better fit for demographic change and more resilient to shocks. Proactive measures—such as identifying people who need care support during emergencies, establishing care protocols to respond to emerging LTC needs, putting in place weather-health alerting, and strengthening caregiver training—must be embedded before events. During crises, ensuring continuity of LTC care is paramount, including by informal providers, as is effective coordination with emergency response services. Recovery should restore LTC services and support rapidly, including through coordination with broader recovery efforts. While examples of such measures are largely drawn from countries with more mature LTC systems, they offer good practices that can be integrated into the provision of specific LTC services and support provided by ministries of social protection or health, no matter how nascent.

References

- AGE. 2024. Insights from DANA Floods: Older People as Key Actors in Fostering Resilience Against Climate Disasters. <https://www.age-platform.eu/insights-from-dana-floods-older-people-as-key-actors-in-fostering-resilience-against-climate-disasters/>
- ADB (Asian Development Bank). 2022. By the Numbers: Climate Change in Central Asia. Manila. <https://www.adb.org/news/features/numbers-climate-change-central-asia>
- Ayalon, L., N. Keating, K. Pillemer, and K. Rabheru. 2021. "Climate Change and Mental Health of Older Persons: A Human Rights Imperative." *The American Journal of Geriatric Psychiatry*, 29(10): 1038-1040. <https://doi.org/10.1016/j.jagp.2021.06.015>
- Ballester, J., M. Quijal-Zamorano, R. Fernando Méndez Turrubiates, F. Pegenaute, F.R. Herrmann, J.M. Robine, X. Basagaña, C. Tonne, J.M. Antó, and H. Achebak. 2023. "Heat-related mortality in Europe during the summer of 2022." *Nature Medicine*, 29(8): 1905–1914. <https://doi.org/10.1038/s41591-023-02419-z>
- Bednar-Friedl, B., R. Biesbroek, D.N. Schmidt, P. Alexander, K.Y. Børsheim, J. Carnicer, E. Georgopoulou, M. Haasnoot, G. Le Cozannet, P. Lionello, O. Lipka, C. Möllmann, V. Muccione, T. Mustonen, D. Piepenburg, and L. Whitmarsh. 2022: Europe. In: *Climate Change 2022: Impacts, Adaptation and Vulnerability*. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press, Cambridge, UK and New York, NY, USA, pp. 1817–1927. <https://doi.org/10.1017/9781009325844.015>
- Benhamou, S., and J. Flamand. 2023. Policy Brief: Work in the Climate Change Challenge. Paris: Haut-commissariat à la Stratégie et au Plan.
- Bogovič, P., S. Lotrič-Furlan, T. Avšič-Županc, L. Lusa, and F. Strle. 2018. "Factors associated with severity of tick-borne encephalitis: A prospective observational study." *Travel Medicine and Infectious Disease*, (26): 25–31. <https://doi.org/10.1016/j.tmaid.2018.10.003>
- Bowen, T., C. del Ninno, C. Andrews, S. Coll-Black, U. Gentilini, K. Johnson, Y. Kawasoe, A. Kryeziu, B. Maher, and A. Williams. 2020. Adaptive Social Protection: Building Resilience to Shocks. International Development in Focus. Washington DC: World Bank. <https://hdl.handle.net/10986/33785>
- C40 Cities. 2022. Climate-Resilient Care for Older People in the Context of the Global Green New Deal and Just Transition. https://www.c40knowledgehub.org/s/article/Climate-resilient-care-for-older-people-in-the-context-of-the-Global-Green-New-Deal-and-Just-Transition?language=en_US
- Cabinet Secretariat, Government of Japan. 2014. Fundamental Plan for a National Resilience – Creating a Strong and Resilient Country." Tokyo: Cabinet Secretariat. https://www.cas.go.jp/jp/seisaku/kokudo_kyoujinka/en/fundamental_plan.html
- Cardona, O.D., M.K. van Aalst, J. Birkmann, M. Fordham, G. McGregor, R. Perez, R.S. Pulwarty, E.L.F. Schipper, and B.T. Sinh. 2012. Determinants of risk: exposure and vulnerability. In: *Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation* [Field, C.B., V. Barros, T.F. Stocker, D.

- Qin, D.J. Dokken, K.L. Ebi, M.D. Mastrandrea, K.J. Mach, G.-K. Plattner, S.K. Allen, M. Tignor, and P.M. Midgley (eds.)). A Special Report of Working Groups I and II of the Intergovernmental Panel on Climate Change (IPCC). Cambridge University Press, Cambridge, UK, and New York, NY, USA, pp. 65-108. https://www.ipcc.ch/site/assets/uploads/2018/03/SREX-Chap2_FINAL-1.pdf
- Community of Madrid. 2024. Action Plan for High Temperature Episodes 2024: Specific Intervention Plan for Social and Health Care Centers (Spanish). https://www.comunidad.madrid/sites/default/files/doc/sanidad/calor/plan_actuacion_episodios_altas_temperaturas_2024_02_plan_especifico_intervencion_centros_sociosanitarios.pdf
- C3S and WMO (Copernicus Climate Change Service and World Meteorological Organization). 2025. European State of the Climate 2024. <https://climate.copernicus.eu/esotc/2024> and <http://doi.org/10.24381/14i9-s541>
- De Zwart, F., B. Brunekreef, E. Timmermans, D. Deeg, and U. Gehring. 2018. "Air Pollution and Performance-Based Physical Functioning in Dutch Older Adults." *Environmental Health Perspectives*, 126(1):017009. <https://doi.org/10.1289/EHP2239>
- Del Prete, V., A. Mateo-Urdiales, A. Bueno-Cavanillas, and P. Ferrara. 2019. "Malaria Prevention in the Older Traveller: A Systematic Review." *Journal of Travel Medicine*, 26(7).
- Dutchen, S. 2021. "The Effects of Heat on Older Adults." *Harvard Medicine Magazine*. <https://magazine.hms.harvard.edu/articles/effects-heat-older-adults>
- Downer, B., A. Holland, S. Li, and H. Xu. Winter Storm Uri, Mortality, and Health Care Use of Nursing Home Residents. 2025. *JAMA Network Open*, 8(4):e254111. <https://doi.org/10.1001/jamanetworkopen.2025.4111>
- Du, W., G.J. FitzGerald, M. Clark, and X.Y. Hou. 2010. "Health Impacts of Floods." *Prehospital and Disaster Medicine*, 25(3): 265–272. <https://doi.org/10.1017/S1049023X00008141>
- Elia, M., and R.J. Stratton. 2005. "Geographical inequalities in nutrient status and risk of malnutrition among English people aged 65 y and older." *Nutrition*, 21(11–12): 1100–1106. <https://doi.org/10.1016/j.nut.2005.03.005>
- EU (European Union). 2014. Adequate social protection for long-term care needs in an ageing society. 2014. https://www.sozialpolitik-aktuell.de/files/sozialpolitik-aktuell/Politikfelder/Europa-Internationales/Dokumente/LTC_social_protection_2014.pdf
- Favaro-Moreira, N.C., S. Krausch-Hofmann, C. Matthys, C. Vereecken, E. Vanhauwaert, A. Declercq, G.E. Bekkering, and J. Duyck. 2016. "Risk factors for malnutrition in older adults: A systematic review of the literature based on longitudinal data." *Advances in Nutrition*, 7(3): 507-522. <https://doi.org/10.3945/an.115.011254>
- Ford, P. "Heat Waves: How France Has Cut Death Toll 90% Since 2003." *Christian Science Monitor*, November 4, 2019. <https://www.csmonitor.com/World/Europe/2019/1104/Heat-waves-How-France-has-cut-death-toll-90-since-2003>

- Fried, L., C.M. Tangen, J. Walston, A.J. Newman, C. Hirsch, J. Gottdiener, T. Seeman, R. Tracy, W. Kop, G. Burke, and M. McBurnie. 2001. "Frailty in Older Adults: Evidence for a Phenotype." *The Journals of Gerontology: Series A*, 56(3): M146-M157.
- Gavazzi, G., and K.-H. Krause. 2002. "Aging and infection." *The Lancet*, 2(11): 659-666. [https://doi.org/10.1016/s1473-3099\(02\)00437-1](https://doi.org/10.1016/s1473-3099(02)00437-1)
- Ghahramani, A., P. Galicia, D. Lehrer, Z. Varghese, Z. Wang, and Y. Patel. 2020. "Artificial Intelligence for Efficient Thermal Comfort Systems: Requirements, Current Applications and Future Directions." *Frontiers in Built Environment*, 6. <https://doi.org/10.3389/fbuil.2020.00049>
- Government of France. 2014. "Recommandations sanitaires du Plan national canicule." https://sante.gouv.fr/IMG/pdf/Recommandations_sanitaires_Plan_canicule_2014.pdf
- Government of France. 2024. "The right steps to take to protect yourself and your loved ones in case of extreme heat." <https://www.service-public.fr/particuliers/actualites/A14978?lang=en>
- Government of Spain. 2025. "Instituto de Mayores y Servicios Sociales." <https://imserso.es/en/>
- GuideStar Eldercare. "Dehydration, Cognitive Functioning, & ADLs." Tuesday May 14, 2024. <https://guidestareldercare.com/blog/dehydration-cognitive-functioning-adls>
- Gutman, G., P. Gorr, and J. Swan. 2010. "The role of technology in emergency preparedness and response for seniors: Lessons learned and opportunities for innovation." *Gerontechnology*, 9(2). <http://dx.doi.org/10.4017/gt.2010.09.02.041.00>
- Hajat, S., M. O'Connor, and T. Kosatsky. 2010. "Health Effects of Hot Weather: From Awareness of Risk Factors to Effective Health Protection." *The Lancet*, 375(9717): 856-863. [https://doi.org/10.1016/s0140-6736\(09\)61711-6](https://doi.org/10.1016/s0140-6736(09)61711-6)
- Hallegatte, S., M. Bangalore, L. Bonzanigo, M. Fay, T. Kane, U. Narloch, J. Rozenberg, D. Treguer, and F. Vogt-Schilb. 2016. *Shock Waves: Managing the Impacts of Climate Change on Poverty*. Climate Change and Development Series. Washington, DC: World Bank. <https://hdl.handle.net/10986/22787>
- Hansen, M., S. Samannodi, R. Castelblanco, and R. Hasbun. 2020. "Clinical epidemiology, risk factors, and outcomes of encephalitis in older adults." *Clinical Infectious Diseases*, 70(11): 2377–2385. <https://doi.org/10.1093/cid/ciz635>
- Hill, R., T. Nguyen, and M.K. Doan. 2024. *Climate and equity: A framework to understand welfare impacts and guide policy action*. World Bank. <https://hdl.handle.net/10986/42259>
- Honda, Y., M. Kondo, G. McGregor, H. Kim, Y.-L. Guo, Y. Hijioka, M. Yoshikawa, K. Oka, S. Takano, S. Hales, and R.S. Kovats. 2013. "Heat-related mortality risk model for climate change impact projection." *Environmental Health and Preventive Medicine*, 19(1): 56-63. <https://doi.org/10.1007/s12199-013-0354-6>
- Hu, H., B. Li, H. Hikichi, I. Kawachi, and X. Li. 2024. "Long-term trajectories of cognitive disability among older adults following a major disaster." *JAMA Network Open*, 7(12): e2448277. <https://doi.org/10.1001/jamanetworkopen.2024.48277>

IPCC (Intergovernmental Panel on Climate Change). 2023. Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. <https://www.ipcc.ch/report/sixth-assessment-report-cycle/>

Kosanica, A., J. Petzold, B. Martin-Lopez, and M. Razanajatovo. 2022. "An inclusive future: disabled populations in the context of climate and environmental change." *Current Opinion in Environmental Sustainability*, 55: 101159. <https://doi.org/10.1016/j.cosust.2022.101159>

Kryspin-Watson, J., J. Pollner, and S. Nieuwejaar. 2008. Climate Change Adaptation in Europe and Central Asia: Disaster Risk Management. Washington DC: World Bank. <http://hdl.handle.net/10986/25985>

Le, G.E., P.N. Breyse, A. McDermott, S.E. Eftim, A. Geyh, J.D. Berman, and F.C. Curriero. 2014. "Canadian forest fires and the effects of long-range transboundary air pollution on hospitalizations among the elderly." *ISPRS International Journal of Geo-Information*, 3(2): 713–731. <https://doi.org/10.3390/ijgi3020713>

Liu, Z., J.P. Murphy, R. Maghirang, and D. Devlin. 2016. "Health and Environmental Impacts of Smoke From Vegetation Fires: A Review." *Journal of Environmental Protection*, 7: 1860–1885. <http://dx.doi.org/10.4236/jep.2016.712148>

Madani Hosseini, M., M. Zargoush, S. Ghazalbash. 2024. "Climate crisis risks to elderly health: strategies for effective promotion and response." *Health Promotion International*, 39(2). <https://doi.org/10.1093/heapro/daae031>

Madani Hosseini, M., S. Beheshti, J. Heydari, M. Zangabadi, and M. Zargoush. 2024a. "Elderly care facility location in the face of the climate crisis: A case study in Canada." *International Journal of Disaster Risk Reduction*. 108: 104516. <https://www.sciencedirect.com/science/article/pii/S2212420924002784>

Maddocks, A., R.S. Young, and P. Reig. 2015. "Ranking the World's Most Water-Stressed Countries in 2040." World Resources Institute. <https://www.wri.org/insights/ranking-worlds-most-water-stressed-countries-2040>

Magan, G. "Can We Protect Older Adults from Climate Disasters?" *The LeadingAge LTSS Center @UMass Boston*, July 3, 2024. <https://www.ltsscenter.org/can-we-protect-older-adults-from-climate-disasters/>

MDPH (Massachusetts Department of Public Health). 2025. Learn About Climate Hazards. Commonwealth of Massachusetts, United States. <https://www.mass.gov/info-details/learn-about-climate-hazards>

MDPH (Massachusetts Department of Public Health). 2025a. Boston, Massachusetts. Extreme Heat and Poor Air Quality. <https://www.mass.gov/info-details/extreme-heat-and-poor-air-quality>

McBrien, H., S. Rowland, T. Benmarhnia, S.Y. Tartof, B. Steiger, and J.A. Casey. 2023. "Wildfire exposure and health care use among people who use durable medical equipment in Southern California." *Epidemiology*, 34(5): 700-711. <https://doi.org/10.1097/ede.0000000000001634>

Ministry of the Environment, Government of Japan. 2025. WBGT Heat Stress Index. Tokyo: Ministry of the Environment. <https://www.wbgt.env.go.jp/en/alert.php>

Ministry of Health, Government of Spain. 2024. National Action Plan for Preventive Measures against Excessive Temperatures over Health 2024. Madrid: Ministry of Health. https://www.sanidad.gob.es/areas/sanidadAmbiental/riesgosAmbientales/calorExtremo/publicaciones/docs/planNacionalExcesoTemperaturas_2024.pdf

Minnesota Department of Health. 2023. Long Term Care Preparedness Toolkit. St. Paul, Minnesota. <https://www.health.state.mn.us/communities/ep/ltc/baseplan.pdf>

Mitchell, D., Y.T.E. Lo, E. Ball, J.L. Godwin, O. Andrews, R. Barciela, L.B. Ford, C. Di Napoli, K.L. Ebi, N.S. Fucker, A. Gasparrini, B. Golding, C.L. Gregson, G.J. Griffith, S. Khalid, C. Robinson, D.N. Schmidt, C.H. Simpson, R.S.J. Sparks, and J.G. Walker. 2024. "Expert Judgement Reveals Current and Emerging UK Climate-Mortality Burden." *The Lancet: Planetary Health*, 8(9): 684-694. [https://doi.org/10.1016/s2542-5196\(24\)00175-x](https://doi.org/10.1016/s2542-5196(24)00175-x)

National Council on Disability. 2023. "The Impacts of Extreme Weather Events on People with Disabilities." Washington DC. <https://www.ncd.gov/assets/uploads/reports/2023/ncd-extreme-weather-2023.pdf>

Naumann, G., L. Alfieri, K. Wyser, L. Mentaschi, R.A. Betts, H. Carrao, and L. Feyen. 2018. "Global Changes in Drought Conditions Under Different Levels of Warming." *Geophysical Research Letters*, 45(7): 3285–3296. <https://doi.org/10.1002/2017GL076521>

Nazri, N.S., D. Vanoh, and S.K. Leng. 2021. "Malnutrition, IOW Diet Quality and Its Risk Factors Among Older Adults with Low Socio-Economic Status: A Scoping Review." *Nutrition Research Reviews*, 34(1): 107116. <https://doi.org/10.1017/s0954422420000189>

OHCHR (Office of the United Nations High Commissioner for Human Rights). 2023. Care and Support Is About Human Rights. Geneva. <https://www.ohchr.org/en/stories/2023/10/care-and-support-about-human-rights>

OECD (Organisation for Economic Co-operation and Development). 2025. "Long-Term Care Resources and Utilisation." *OECD Health Statistics 2019. Definitions, Sources and Methods: January 5, 2025*. <http://stats.oecd.org/wbos/fileview2.aspx?IDFile=a29d8d56-e227-4e7b-a90f-b2232b074723>

Okuda, H., and J. Tomio. 2024. Disaster Preparedness for Aging Populations: Lessons from Japan. AHWIN PAPERS: Lessons for Aging-Related Policy. No. 4.

Pascal, M., V. Wagner, A. Le Tertre, K. Laaidi, C. Honore, F. Benichou, and P. Beaudet. 2010. "Definition of Temperature Thresholds: The Example of the French Heat Wave Warning System." *International Journal of Biometeorology*, 54(1): 75–84.

Pascal, M., V. Wagner, M. Corso, K. Laaidi, A. Ung, and P. Beaudet. 2018. "Heat and cold related-mortality in 18 French cities." *Environment International*, 121: 189-198. <https://doi.org/10.1016/j.envint.2018.08.049>

Prina, M., N. Khan, S.A. Khan, J. Castro Caicedo, A. Peycheva, V. Seo, S. Xue. and R. Sadana. 2024. "Climate change and healthy ageing: An assessment of the impact of climate hazards on older people." *Journal of Global Health*, 14: 04101. <https://doi.org/10.7189/jogh.14.04101>

- Pruchno, R., M. Wilson-Genderson, A., Heid, and F. Cartwright. 2019. Type of Disaster Exposure Affects Functional Limitations of Older People Six Years Later. *The Journals of Gerontology Series A Biological Sciences and Medical Sciences* 75(11). DOI:10.1093/gerona/glz258
- Ratwatte, P., H. Wehling, S. Kovats, O. Landeg, and D. Weston. 2022. "Factors Associated with Older Adults' Perception of Health Risks of Hot and Cold Weather Event Exposure: A Scoping Review." *Frontiers in Public Health*, 10:939859. <https://doi.org/10.3389/fpubh.2022.939859>
- Regional Environmental Centre for the Caucasus. 2013. Views from the Frontline: Country Report: Georgia. Tbilisi. <https://rec-caucasus.org/wp-content/uploads/2020/08/1557345410.pdf>
- Roberts, T., V. Seymour, K. Brooks, R. Thompson, C. Petrokofsky, E. O'Connell, and O. Landeg. 2022. "Stakeholder perspectives on extreme hot and cold weather alerts in England and the proposed move towards an impact-based approach" *Environmental Science and Policy*, 136: 467-475. <https://doi.org/10.1016/j.envsci.2022.07.012>
- Robine, J.M., S.L.K. Cheung, S. Le Roy, H. Van Oyen, C. Griffiths, J.P. Michel and F.R. Herrmann. 2008. "Death Toll Exceeded 70,000 in Europe during the Summer of 2003." *Comptes Rendus Biologies*, 331(2): 171–178. <https://doi.org/10.1016/j.crv.2007.12.001>
- Shaw, R., Y. Luo, T.S. Cheong, S. Abdul Halim, S. Chaturvedi, M. Hashizume, G.E. Insarov, Y. Ishikawa, M. Jafari, A. Kitoh, J. Pulhin, C. Singh, K. Vasant, and Z. Zhang, 2022: Asia. In: *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press, Cambridge, UK and New York, NY, USA, pp. 1457–1579. <https://doi.org/10.1017/9781009325844.012>
- Simpson, N., K. Mach, A. Constable, J. Hess, R. Hogarth, M. Howden, J. Lawrence, R. Lempert, V. Muccione, B. Mackey, M. New, B. O'Neill, F. Otto, H. Poertner, A. Reisinger, D. Roberts, D. Schmidt, S. Senevieratne, S. Strongin, M. van Aalst, and C. Trisos. 2021. "A Framework for Complex Climate Change Risk Assessment." *One Earth*, 4(4): 489–501. <https://doi.org/10.1016/j.oneear.2021.03.005>
- Sinnott, E., M. Bussolo, and J. Koettl. 2015. Golden Aging: Prospects for Healthy, Active, and Prosperous Aging in Europe and Central Asia. Washington DC: World Bank. <http://hdl.handle.net/10986/22018>
- Spasova, S., R. Baeten, and B. Vanhercke. 2018. "Challenges in Long-Term Care in Europe." *Eurohealth Observer*. <https://iris.who.int/bitstream/handle/10665/332533/Eurohealth-24-4-7-12-eng.pdf>
- Stanke, C., M. Kerac, C. Prudhomme, J.M. Medlock, and V. Murray. 2013. "Health Effects of Drought: A Systematic Review of the Evidence." *PLoS Currents*, 5. <https://doi.org/10.1371/currents.dis.7a2cee9e980f91ad7697b570bcc4b004>
- Stein, P., and M. Stein. 2022. "Climate change and the right to health of people with disabilities." *The Lancet*, 10(1): e24-e25. [https://www.thelancet.com/journals/langlo/article/PIIS2214-109X\(21\)00542-8/fulltext](https://www.thelancet.com/journals/langlo/article/PIIS2214-109X(21)00542-8/fulltext)

Triantafyllou, J., M. Naiditch, K. Repkova, S. Carretero, T. Emilsson, R. Bednarik, L. Brichtova, F. Ceruzzi, L. Cordero, K. Mingot, J. Ritter, and K. Leichsenring. 2010. "Informal Care in the Long-term Care System: European Overview Paper." *WP5, European Centre for Social Welfare Policy and Research*, Athens/Vienna.

UKHSA (United Kingdom Health Security Agency). 2023. Supporting Vulnerable People Before and During Cold Weather: For Adult Social Care Managers. <https://www.gov.uk/government/publications/cold-weather-and-health-supporting-vulnerable-people/supporting-vulnerable-people-before-and-during-cold-weather-for-adult-social-care-managers>

UKHSA (United Kingdom Health Security Agency). 2024. Heat-Health Alerting System. <https://www.gov.uk/guidance/heat-health-alerting-system>

UKHSA (United Kingdom Health Security Agency). 2024a. Heat-Health Alerting System: Guidance for Health and Social Care Providers. <https://www.gov.uk/guidance/heat-health-alerting-system-guidance-for-health-and-social-care-providers>

UKHSA (United Kingdom Health Security Agency). 2024b. Top Tips for Keeping Warm and Well This Winter. <https://www.gov.uk/government/publications/keep-warm-keep-well-leaflet-gives-advice-on-staying-healthy-in-cold-weather/top-tips-for-keeping-warm-and-well-this-winter>

UKHSA (United Kingdom Health Security Agency). 2025. Heat-Health Alert Action Card for Health and Social Care Providers. <https://www.gov.uk/guidance/heat-health-alert-action-card-for-health-and-social-care-providers>

UN (United Nations). 2021. Decade of Healthy Ageing: Plan of Action, 2021-2030. <https://social.desa.un.org/sdn/decade-of-healthy-ageing-2021-2030#:~:text=The%20United%20Nations%20Decade%20of,of%20older%20people%2C%20their%20families%2C>

UNCRPD (United Nations Convention on the Rights of Persons with Disabilities). 2006. Convention on the Rights of Persons with Disabilities and Optional Protocol. <https://www.un.org/disabilities/documents/convention/convoptprot-e.pdf>

UNICEF (United Nations Children's Fund). 2025. Global Disability Inclusion Report: Accelerating Disability Inclusion in a Changing and Diverse World. https://www.globaldisabilitysummit.org/wp-content/uploads/2025/03/GIP03351-UNICEF-GDIR-Full-report_Proof-4.pdf

UNFCCC (United Nations Framework Convention on Climate Change). 2012. Technical Paper: Slow Onset Events. FCCC/TP/2012/7. <https://unfccc.int/resource/docs/2012/tp/07.pdf>

United States Environmental Protection Agency. 2025. Climate Change and Health: Older Adults. Washington DC. <https://www.epa.gov/climateimpacts/climate-change-and-health-older-adults>

Warshaw, G., and E. Bragg. 2014. "Preparing the Health Care Workforce to Care for Adults with Alzheimer's Disease and Related Dementias." *Health Affairs* 33(4): 633–41. <https://doi.org/10.1377/hlthaff.2013.1232>

- Wilbur, J., D. Ruuska, S. Nawaz, and J. Natukunda. 20214. *Climate Risks to Water, Sanitation and Hygiene Services and Evidence of Inclusive and Effective Interventions in Low and Middle-Income Countries: A Scoping Review*. medRxiv. <http://dx.doi.org/10.1101/2024.08.21.24312122>
- Willner, S.N., A. Levermann, F. Zhao, and K. Frieler. 2018. "Adaptation Required to Preserve Future High-End River Flood Risk at Present Levels." *Science Advances*, 4(1). <https://doi.org/10.1126/sciadv.aao1914>
- Wollschlaeger, S., A. Sadhu, G. Ebrahimi, and A. Woo. 2022. "Investigation of Climate Change Impacts on Long-Term Care Facility Occupants." *City and Environment Interactions*, 13:100077. <https://doi.org/10.1016/j.cacint.2021.100077>
- World Bank. The ABCs of Long-Term Care: A Functional Framework for Building System Capacities in Europe and Central Asia. Forthcoming.
- World Bank. 2010. Long Term Care Policies for Older Populations in New EU Member States and Croatia: Challenges and Opportunities. Washington DC. <https://hdl.handle.net/10986/12898>
- World Bank Group. 2021. Climate Risk Country Profile: Albania. Washington DC. https://climateknowledgeportal.worldbank.org/sites/default/files/2021-06/15765-WB_Albania%20Country%20Profile-WEB.pdf
- World Bank Group. 2021a. Climate Risk Country Profile: Armenia. Washington DC. https://climateknowledgeportal.worldbank.org/sites/default/files/2021-06/15765-WB_Armenia%20Country%20Profile-WEB_0.pdf
- World Bank Group. 2021b. Climate Risk Country Profile: Georgia. Washington DC. https://climateknowledgeportal.worldbank.org/sites/default/files/2021-06/15836-WB_Georgia%20Country%20Profile-WEB.pdf
- World Bank Group. 2021c. Climate Risk Country Profile: Kazakhstan. Washington DC. https://climateknowledgeportal.worldbank.org/sites/default/files/2021-08/15834-WB_Kazakhstan%20Country%20Profile-WEB.pdf
- World Bank Group. 2021d. Climate Risk Country Profile: Tajikistan. Washington DC. https://climateknowledgeportal.worldbank.org/sites/default/files/2021-09/15919-WB_Tajikistan%20Country%20Profile-WEB.pdf
- World Bank Group. 2021e. Climate Risk Country Profile: Uzbekistan. Washington DC. https://climateknowledgeportal.worldbank.org/sites/default/files/2021-09/15838-WB_Uzbekistan%20Country%20Profile-WEB.pdf
- World Bank. 2022. Emergency Preparedness and Response (EP&R): Case Study of Japan. Washington, D.C.: World Bank Group.
- World Bank. 2024. Preparing Health Systems for Shocks: Japan's Experience of Enhancing the Resilience of its Health System. Washington, D.C.: World Bank Group.
- World Bank. 2025. Unlivable: Extreme Heat in Cities in Europe and Central Asia. Washington DC.

- WHO (World Health Organization). 2015. World Report on Ageing and Health. Geneva. <https://www.who.int/publications/i/item/9789241565042>
- WHO (World Health Organization). 2015a. Heat-Waves and Health: Guidance on Warning-System Development. Geneva. <https://cdn.who.int/media/docs/default-source/climate-change/heat-waves-and-health---guidance-on-warning-system-development.pdf>
- WHO (World Health Organization). 2020. World Health Statistics 2020: Monitoring Health for the SDGs. Geneva. <https://www.who.int/publications/i/item/9789240005105>
- WHO (World Health Organization). 2023. Drinking-water. Geneva. <https://www.who.int/news-room/fact-sheets/detail/drinking-water>
- WMO (World Meteorological Organization). 2025. Extreme Weather. Geneva. <https://wmo.int/topics/extreme-weather>
- Yoneda, K., S. Hosomi, H. Ito, Y. Togami, S. Oda, H. Matsumoto, J. Shimazaki, and H. Ogura. 2024. "How can heatstroke damage the brain? A mini review." *Frontiers In Neuroscience*, 18. <https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2024.1437216/full>
- Yotsui, M., C. Campbell, and T. Honma. 2016. "Collective Action by Older People in Natural Disasters: The Great East Japan Earthquake." *Ageing & Society*, 36(5): 1052–1082. <http://dx.doi.org/10.1017/S0144686X15000136>
- Young, S., and A.M. Bergseng. 2020. Delivering social care in a changing climate. University of Edinburgh, ClimateXChange. Edin2020.i2020.d2020.E2020. <https://www.climateexchange.org.uk/wp-content/uploads/2023/09/cxc-delivering-social-care-in-a-changing-climate-june-2020.pdf>
- Zhou, W., Q. Wang, R. Li, Z. Zhang, W. Wang, F. Zhou, and L. Ling. 2023. "The effects of heatwave on cognitive impairment among older adults: Exploring the combined effects of air pollution and green space." *Science of The Total Environment*, 904. <https://www.sciencedirect.com/journal/science-of-the-total-environment/vol/904/suppl/Chttps://doi.org/10.1016/j.scitotenv.2023.166534>

Social Protection & Jobs Discussion Paper Series Titles FY 2026

<u>No.</u>	<u>Title</u>
2604	The heat is on: How can long-term care systems in Europe and Central Asia promote climate adaptation?
2603	Rethinking Poverty Reduction: Building an Effective Safety Net in Cameroon
2602	Supporting People and Workers in an Inclusive Green Transition: Framing Social Protection and Labor Policies to Enable and Ensure a Just Green Transition
2601	Refugees and social assistance in low- and middle-income countries: A review of operational experiences

To view Social Protection & Jobs Discussion Papers published prior to 2021, please visit www.worldbank.org/sp.

ABSTRACT

Countries in Europe and Central Asia are experiencing population aging and increasing climate hazards, albeit with local variation in prevalence, intensity, and frequency. Older people and those with functional limitations are disproportionately affected by climate hazards, making long-term care (LTC) systems a natural platform for climate adaptation. Drawing on OECD experiences and global evidence, this paper identifies measures that spans the functions of an LTC system, which ECA countries could adopt before, during, and after a climate hazard to maintain continuity of LTC services and support, respond to climate-induced needs among LTC beneficiaries, and reach affected populations beyond formal LTC provision. Many of these actions involve adapting how LTC services are organized and delivered rather than creating new standalone services.

JEL CODES

I18, Q54, H75, J14, I38.

KEYWORDS

climate change, climate adaptation, long-term care, older adults, people with care needs, people with disabilities

